



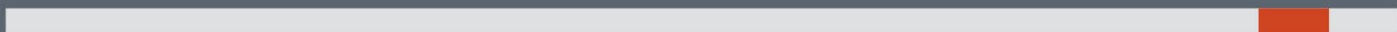
Australian Government

Defence



Defence Infrastructure Engineering Manual

April 2026





Australian Government

Defence

INFRASTRUCTURE ENGINEERING MANUAL

This document is issued for use by Assistant Secretary Environment and Engineering and is effective forthwith.

A handwritten signature in black ink, appearing to read 'Felix B.-Derrick'.

Felix Bowman-Derrick

Assistant Secretary Environment and Engineering

Department of Defence

CANBERRA ACT 2600

23 April 2026

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Infrastructure Engineering Manual

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Assistant Secretary Environment and Engineering Branch

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Environment and Engineering Branch

¹ <https://www.legislation.gov.au/Series/C1968A00063>

² <https://www.legislation.gov.au/Series/C2004A04868>

³ <https://www.legislation.gov.au/Series/C2004A03712>

⁴ <http://drnet/AssociateSecretary/security/policy/Pages/dspf.aspx>

FOREWORD

The Defence Infrastructure Engineering Manual (IEM) describes the mandatory compliance and conformance requirements for design, construction and maintenance of Defence infrastructure. This work is undertaken for the protection of Defence personnel and assets in support of Australian Defence objectives. The expected outcomes are that Defence infrastructure will be designed, constructed, altered and maintained following appropriate certification processes and in compliance with the requirements of the AS/NZ standards, the National Construction Code (NCC), relevant state and territory regulations and all other Defence policies applicable to infrastructure engineering works. The objective of this manual is to reinforce the requirement that works on the Defence estate and other Defence occupied buildings must comply with all relevant Defence policies and align the administrative requirements outlined above as much as practicable.

Requirements

The Secretary and the Chief of the Defence Force (CDF) prescribe the Defence framework for Infrastructure Engineering in this manual.

The Secretary and CDF require Defence personnel, contractors, consultants and outsourced service providers to comply with the requirements of the following:

- a. the NCC including Building Code of Australia, Volumes One and Two and Plumbing Code of Australia Volume Three
- b. applicable Defence policies related to infrastructure engineering work
- c. applicable state and territories regulations.

The IEM also provides the procedural framework for involving appropriately skilled practitioners and prescribed levels of oversight to meet Defence's regulatory obligations.

Key roles, functions and responsibilities

The Assistant Secretary Environment and Engineering (ASEE) is authorised to approve amendments to this manual and associated codes and standards.

The Director of Estate Engineering Policy is responsible for the administration of this policy. This includes responsibility for ensuring the manual remains current and monitoring the implementation of the requirements described herein. Any queries or feedback related to application of this manual shall be submitted to deep.advice@defence.gov.au.

The delivery authority is responsible for the implementation of the requirements of this manual and associated codes and standards.

Definitions and related documents

Chapter 9 contains the glossary and acronyms. The definitions in the glossary are specific to this manual and may not align with Australian Defence glossary definitions.

Review of the manual

The manual will be reviewed 5 years from its date of issue or sooner if necessitated by business requirements.

Minor amendments may be made annually, if required, urgent amendments may also occur.

AMENDMENTS

Proposals for amendment may be sent to:

deep.advice@defence.gov.au

The IEM replaces the following documents:

- Building Works Manual (2020)
- Manual of Fire Protection Engineering, Edition 4, Amendment 4.1 (2023)
- Manual for Infrastructure Engineering Electrical, Edition 2 (2022)
- Smart Infrastructure Handbook (2019)
- Climate Adaptation Handbook (2019)
- Building Energy Performance Guide (2022)

Amendment number	Amendment	Amendment type	Effective date

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CHAPTER 1

APPLICATION

Background and context

- 1.1 The requirements of this manual and its guidelines apply across the Defence estate:
 - a. where new facilities or infrastructure are to be constructed
 - b. where existing facilities or infrastructure are to be refurbished, altered or additions made
 - c. in the operation of existing installations and infrastructure
 - d. in the maintenance of existing installations and infrastructure.
- 1.2 The following are considered Defence infrastructure and the requirements of this manual shall be applied:
 - a. properties leased by Defence in Australia
 - b. facilities delivered and/or managed under Public Private Partnership arrangements
 - c. joint use facilities located on Defence establishments in Australia
 - d. foreign facilities located on Defence establishments in Australia.
- 1.3 Where Defence lease facilities to a third party, the requirements of this manual shall be applied.
- 1.4 It is recommended that when facilities are designed outside of Australia, project leads engage Australian design assistance at an early stage to ensure the installation will meet applicable Australian legislation and regulations, including mandatory Australian standards.
- 1.5 Facilities located overseas shall comply fully with local statutory requirements, standards, and codes. Capability managers or project sponsors may elect to apply the requirements of this manual, in part or in full, to overseas bases or facilities as they consider appropriate.
- 1.6 When works of a temporary nature are undertaken, the specific requirements of this manual may not be applicable. A lower standard can be applied to temporary works required during the period of construction of a project where:
 - a. this offers significant cost or time savings
 - b. the temporary works meet the applicable statutory standards
 - c. it is accepted by the Defence Operating Authority for the Defence electrical distribution networks and systems.
- 1.7 Temporary works are to be removed by the completion of the project. Where the requirements of this manual are not to be applied to any temporary facilities or works, the Project Delivery Authority shall fully document the justification for not applying the requirements of this manual.

Retrospective application of this manual

- 1.8 This manual is applicable to new works and installations and to the alteration, addition or augmentation of existing installations as identified in further chapters of this manual. There is no requirement to apply this manual retrospectively to existing installations which are not otherwise being upgraded or changed. In the event of differing guidance being provided in this manual and in previous infrastructure engineering policy documents, seek clarification of the correct policy from the Directorate of Estate Engineering Policy (DEEP).

CHAPTER 2

GENERAL CONCEPTS

Policy hierarchy

- 2.1 This manual fits in the hierarchy of application set out below. Where there is a conflict between the levels, the more stringent requirement shall apply. This may include state, territory or local government requirements.

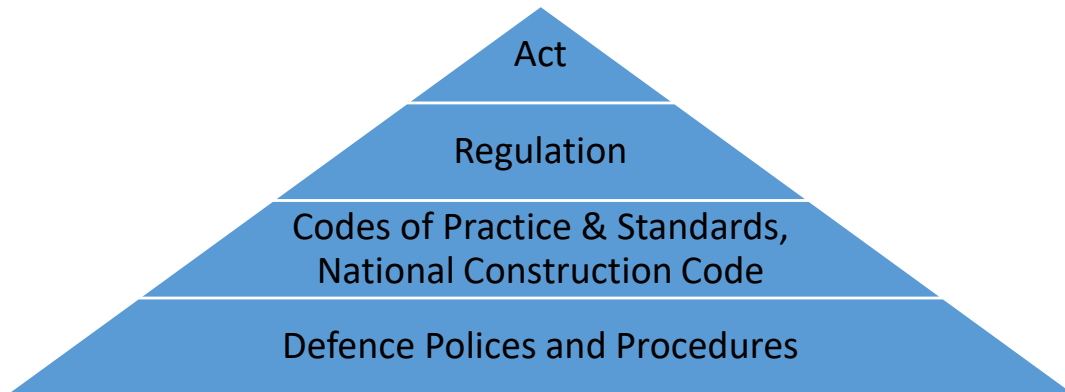


Figure 2.1: Hierarchy of acts, regulations, standards and Defence policies and procedures

Whole of life optimisation

- 2.2 The design, construction, operation and maintenance of Defence infrastructure, assets and installations shall optimise whole-of-life outcomes for Defence. This must consider cost, greenhouse gas emissions and resilience. In determining suitable design, materials and methods the following shall be considered:
- a. the Defence capability supported by the asset and its importance in supporting that capability
 - b. the reliability requirements and the consequences of failure of the asset
 - c. the expected lifetime of the asset in Defence ownership, considering that Defence often own assets from construction through to decommissioning
 - d. the diversity of supply, particularly for embedded generation options
 - e. the expected time between upgrades or refurbishment
 - f. the climatic, environment and service requirements of the area in which the equipment is installed
 - g. availability of maintenance and other support at the location in which the asset is located
 - h. the required survivability of the asset following events such as natural disaster
 - i. sustainable use of resources, energy and water are critical requirements for Defence to maintain and enhance its operations and capability.
- 2.3 For most assets, the above requires the use of good quality commercial or industrial grade materials and practice.

Defence Net Zero and Defence Transition to Clean Energy

- 2.4 The Australian Government, through the *Climate Change Act 2022*, legislated emissions reduction targets. Australia's interim target is a 43% reduction on 2005 emission levels by 2030 and net zero by 2050. Defence is committed to meeting these targets.
- 2.5 As the Australian Government's largest landholder and energy user, Defence has a significant role to play. The National Defence Strategy 2026 (NDS 2026) committed Defence to accelerating its transition to clean energy.
- 2.6 Through the Net Zero in Government Operations Strategy, administered by the Department of Finance, all Australian Government agencies are required to develop an emissions reduction plan. The Defence Net Zero Strategy is Defence's emission reduction plan.
- 2.7 Design decisions made now will affect Defence's emissions and resilience for decades into the future. Adaptation for climate and energy resilience comes in many forms and supports Defence assets and infrastructure to enable operations into the future. The Defence Environment Manual and its associated guidance covers climate resilience in more detail.
- 2.8 This manual provides the technical framework to design infrastructure that meets Defence's operational needs while improving resilience, sustainability, and adaptability. Every engineering decision from materials to systems and technologies affects emissions, energy resilience, and long-term sustainability.
- 2.9 Integrating renewable energy, smart systems, and efficient design strengthens both environmental performance and the self-sufficiency of Defence facilities, especially in our remote and northern bases. Energy resilience is a core infrastructure requirement and a matter of national security.

Reduction of gas usage across the estate

- 2.10 Defence is committed to reducing the use of gas and shifting to electrification in line with the Australian Government policy and the Defence Net Zero Strategy, where this does not compromise accelerated preparedness, operational effectiveness or capability.
- 2.11 From 1 July 2028, new gas appliances (such as for cooking, heating and hot water) or connections should be avoided and at end of life, and where practicable, gas appliances should be replaced with electric alternatives. The asset replacement lifecycle should plan for the electrification of buildings where practical.

Contribution Factor

- 2.12 Defence buildings are categorised in accordance with their contribution to capability and assigned a contribution factor. Guideline 2000 provides detailed guidance on how contribution factors are assessed and managed.

Dispensations

- 2.13 Where non-compliance to a Defence specific policy outlined in this manual is identified, a dispensation will be required. A dispensation cannot be used to provide relief from the requirements of legislation, regulations, or codes or standards.
- 2.14 The dispensation process is detailed in Guideline 2001.

Performance solutions

- 2.15 A performance solution is a method of complying with the National Construction Code that is not based on the prescriptive Deemed-to-Satisfy provisions. The performance solution process is detailed in Guideline 2002.

Alternate conformance solutions & compliance by specific design

- 2.16 Conformance with Defence policy for electrical installations may not be practicable due to technical, site or other project constraints. Any non-conformance with the Defence policy requirements must be addressed via a request for alternative conformance solution as detailed in Guideline 2003.
- 2.17 Certain electrical installations, or portions of electrical installations, because of their unusual requirements, application or intended use, may not fully meet the requirements of Part 2 of AS/NZS 3000. In these circumstances the compliance by specific design process detailed in Guideline 2004 may be used.

Cyber worthiness

- 2.18 Defence Estate Engineering Networks shall withstand cyber security attacks in accordance with Defence and Australian Government policies, in particular the Information Security Manual and the Defence Security Principles Framework. The design and construction of the Engineering Services Network and the systems running on the network shall prevent unauthorised access, including remote access, to the system that could result in interference, operation, monitoring or tampering.

Defence specific requirements

- 2.19 Defence utilises systems and equipment used through broader industry in a manner consistent with commercial and industrial applications. Given the nature of Defence objectives, Defence has some particular requirements in relation to the implementation of systems which may not be consistent with uses in other industries.
- 2.20 Defence utilises a number of specialist systems across Defence establishments to support operational capability objectives. These systems are often not commonplace in broader industry and require specialist knowledge in order to safely and successfully design, construct and operate them. These may not have relevant Australian Standards and may require the application of principles-based engineering in order to achieve compliance by specific design with the principles of standards, rather than the deemed to satisfy conditions e.g. 400Hz switchboard etc.
- 2.21 The following guidelines articulate Defence specific requirements relevant to this chapter:
- a. 2000 – Contribution factors
 - b. 2001 – Dispensation process
 - c. 2002 – Performance Solutions
 - d. 2003 – Alternative conformance solutions
 - e. 2004 – Compliance by Specific Design

CHAPTER 3

BUILDING WORKS

GENERAL PRINCIPLE

- 3.1 The building works chapter of this manual specifies the compliance and conformance requirements for design, construction and maintenance of Defence occupied buildings for the protection of personnel and assets in support of Australian Defence objectives.

Rationale

- 3.2 The rationale of this chapter is to:
- a. reinforce the requirement that building work on the Defence estate and other Defence occupied buildings must comply with the NCC and all relevant Defence policies
 - b. align the administrative requirements for building work on the Defence estate to state and territory building legislation and certification requirements as much as practicable.

Expected outcomes

- 3.3 Defence occupied buildings will be designed, constructed, altered and maintained following appropriate certification processes and compliant with the NCC and all other Defence policies applicable to building work.

National Construction Code

- 3.4 The NCC shall be adopted as the minimum standard for building work for Defence estate buildings. The NCC edition that is applicable for any building work on the Defence estate is the edition that is adopted by the Australian Government at the time of building approval. Relevant state and territory appendices of the NCC for building work on the Defence estate shall be applied for Defence buildings, except for the appendices related to Section J of NCC Volume One and Section H.6 of NCC Volume Two. The requirements of Section J of NCC Volume One and Section H6 of NCC Volume Two are applicable in all states and territories.
- 3.5 Where a NCC referenced document, standard or code is referenced in a Defence policy, the year version of the standard or code in the NCC at the time of building approval is applicable.
- 3.6 Unless otherwise specified in this policy, standards or codes referred to by the Defence policy shall be the current version at the time of building approval.
- 3.7 The requirements of the NCC Volume One are met by demonstrating compliance with the governing requirements of Section A and the performance requirements Sections B to G and I to J of the NCC. In relation to performance requirements, compliance is achieved by either developing a performance solution, a deemed-to-satisfy (DTS) solution and/or a combination of performance solutions and DTS solutions.

Special structures

- 3.8 Where the nature of specialist military facilities prevents the reasonable application of NCC performance requirements, a special structure classification may be applied. Chapter 5 and Guideline 5012 detail further requirements for special structures.

Heritage buildings

- 3.9 Some buildings on the Defence estate may have statutory heritage protections, and may be included in a Commonwealth Heritage Listing (CHL) or National Heritage Listing (NHL).
- 3.10 Buildings with statutory heritage protections must be maintained to comply with relevant building standards and codes. Maintenance works that require changes to buildings can occur if reasonably practicable measures are taken to mitigate their impact on the heritage values of the building. This may include best endeavours to match colour schemes or the use of similar materials to the original construction materials. Further advice should be obtained from the relevant Defence Environment and Sustainability (E&S) directorates.
- 3.11 Where work, health and safety (WHS) risks result from unsafe structures or workplaces, heritage considerations do not outweigh Defence's primary obligation to ensure the health and safety of workers.
- 3.12 An identified WHS risk must be assessed by a competent person and appropriate action must be taken. Priority should be given to eliminating risk (e.g. demolition, replacement of structure) and other higher order controls (e.g. refurbishment and bracing) over administrative controls (e.g. regular inspection, notices and training).
- 3.13 Where a building or asset is cited in the NHL or CHL, E&S advice is required. The E&S directorates will escalate to the Directorate of Environment and Heritage Policy Development if necessary. Inclusion within a listed area does not automatically mean a building or asset is cited. Refer to the Environment Manual for further heritage information.

Pedestrian access

- 3.14 For the purpose of compliance with the DTS provisions of D4D3(1) of the NCC, the term 'main points of a pedestrian entry at the allotment boundary' means the closest accessible walkway for pedestrian access to the building from the nearest public road or open space on the Defence site. For these requirements, the road infrastructure on a Defence site may be considered a public road.

Building work and substantial alterations

- 3.15 All building work shall comply with the requirements of this manual.
- 3.16 Where building work constitutes a substantial alteration, the entire building shall be brought into full compliance and conformance with this policy and the NCC.
- 3.17 Where building work does not constitute a substantial alteration, the following applies:
 - a. the new building work, and any other area required by the Premises Standards: Disability (Access to Premises – Buildings) Standards 2010, shall comply with this policy and the NCC
 - b. new building works shall not reduce the safety within the remainder of the building or adjacent buildings
 - c. existing or adjacent buildings shall not adversely impact safety of the new building work.

Demolition

- 3.18 Demolition of a building must have demolition authorisation and be the subject of a building approval. Demolition work must comply with AS 2601 – The Demolition of Structures. Additional requirements may be detailed in the Defence Contamination Management Manual.

Existing buildings – no construction works

- 3.19 Existing buildings that are not the subject of any building work must comply with the requirements of any certificates of completion for the building.

BUILDING CERTIFICATION

- 3.20 This section covers and provides details on:
- a. compliance and conformance requirements
 - b. all building work that must be certified in accordance with this chapter
 - c. requirements of building surveyor
 - d. certification functions required for building work on the Defence estate carried out by a building surveyor.

Competent person assessments

- 3.21 The building surveyor must assess whether each person engaged to prepare building design or specification documentation is a competent person in their relevant area of expertise. The building surveyor must be satisfied of the following matters:
- a. where accreditation is required in the relevant area of practice by the state or territory where the work is to be undertaken, the person being assessed must hold current accreditation
 - b. where accreditation is not required in the relevant practice area by the state or territory where the work is to be undertaken the person being assessed must hold appropriate qualifications and skills as determined by the building surveyor
 - c. in addition to a. or b. above, the person being assessed has appropriate skills and experience having regard to the nature of the building work and buildings that the person will be providing services for.
- 3.22 The building surveyor shall:
- a. confirm, in the building approval that the assessment of each competent person has been undertaken
 - b. document how they determined that each person who provides building design or specification for the building work is a competent person. This documentation must be retained for a minimum period of 10 years from the issue date of the certificate of completion
 - c. provide such records or documents to the Defence policy area on request.

Note: Confirmation that the building surveyor has assessed the person as competent must be appended to all design and installation certification.

Building assessment

- 3.23 Prior to issuing a building approval or any variation to a building approval the surveyor must undertake an assessment.
- 3.24 Each building assessment undertaken by a building surveyor must be documented in a report which shows the status of the proposed designs and specifications with:
- a. the relevant requirements of the NCC

- b. any requirements that apply because of the contribution factor for the building
 - c. any requirement that applies under this policy if the building is/will be a special structure.
- 3.25 The building surveyor shall provide a copy of their assessment report to the policy contact on request. This assessment report must be retained for at least 10 years from the issue date of the certificate of completion.

Conflict of interest and the conduct of building surveyors

- 3.26 A building surveyor cannot carry out certification functions if they or a related person has an actual, potential or perceived conflict of interest in that they:
- a. carry out the building work that is the subject of the building approval process
 - b. have participated in the preparation of the design for all or part of the building work that is the subject of the building approval
 - c. have any pecuniary interest in a company or entity that is engaged in relation to building work that is the subject of the building approval
 - d. have a direct or indirect pecuniary interest in the building work
 - e. are engaged by the builder or designer to perform a function other than a certification function for the project or regulatory advice about any matter
 - f. arrange any third-party approvals e.g. plumbing or electrical to achieve compliance with building approval or certificate of completion requirements.
- 3.27 A related person constitutes:
- a. if the building surveyor is a member of a partnership, another partner in the partnership
 - b. if the building surveyor is a body corporate, a director of the body corporate or any related body corporate of the body corporate an employer, employee or contractor of the building surveyor
 - c. the spouse, domestic partner, sibling, parent or child of the building surveyor or any person with whom the building surveyor has a contractual arrangement that might reasonably be seen to give rise to a conflict between the building surveyor when exercising the certification functions and their interests under the arrangement.
- 3.28 Building surveyors are required to comply with any code of conduct that applies to their accreditation when carrying out certification functions under this IEM.
- 3.29 This chapter does not prevent a building surveyor from being a sub-contractor to a builder engaged by Defence provided conflict of interest management requirements are met.
- 3.30 Where there is an actual, potential or perceived conflict between this document and any applicable code of conduct, a building surveyor should apply to the relevant policy contact for a determination on how the conflict should be resolved.

Building work

- 3.31 All building work must be the subject of a building approval and a certificate of completion except for exempt building work. All exempt building work must be completed in accordance with the applicable design standards and codes, and the builder or installer shall provide certificates or evidence of compliance and conformance as appropriate.

- 3.32 A building approval and certificate of completion can only be issued by the surveyor appointed for the specific building work. The surveyor must carry out all certification functions in relation to that building work. When the building surveyor cannot continue to act for one of the reasons set out below, a replacement surveyor can be sought. The reasons for discontinuance include:
- a. sickness or death
 - b. change of employer or company
 - c. conflict of interest
 - d. no longer in business
 - e. accreditation reduced, suspended or cancelled
 - f. inadequate insurance.

Building approval

- 3.33 A building approval is required for building work that is not exempt in exempt building work.
- 3.34 A building approval is issued by the building surveyor when:
- a. all design and specification documentation and design certificates have been received, and the surveyor is satisfied that the building work can comply with the IEM and the NCC if constructed in accordance with the building approval and its referenced documents
 - b. the building assessment report has been completed, having regard to the Fire Engineering requirements of this manual.
- 3.35 Supporting evidence for a building approval includes but is not limited to:
- a. project requirements documentation
 - b. building contribution factor (Guideline 2000) form, if applicable
 - c. hangar assessment form (Guideline 5005), if applicable
 - d. written agreement from the policy contact for the building assessment plan applicable to the use of a special structure classification
 - e. evidence of consultation with relevant referral agencies or as required under Defence policy, including but not limited to:
 - i. responding fire service
 - ii. plumbing and drainage
 - iii. electrical and gas
 - f. architectural, structural, services, or civil design documentation and design certificates
 - g. specialist practitioners design certificates as appropriate e.g. bushfire, access, energy
 - h. plumbing work and drainage work design documentation and design certificates as appropriate e.g. disposal of non-domestic waste, an increase in the water demand or sewer load, discharge stormwater to sewer
 - i. performance solutions are supported by the policy contact
 - j. dispensations have been approved at Senior Executive Service – Band 1 or equivalent level

- k. demolition authorisation, if applicable
 - l. environmental requirements, if applicable.
- 3.36 Unless set out above, the minimum inclusions required for the stage that the design is at the time the building approval is issued having regard to the Fire Engineering Design Inclusions.
- 3.37 A building approval must:
- a. identify the building location and property details including the Garrison Estate Management System (GEMS) identification
 - b. identify the contribution factor for the building
 - c. identify the aircraft information provided on the hangar assessment form for an aircraft hangar, if applicable
 - d. identify the NCC and policy versions applicable to the building approval
 - e. identify whether a special structure classification is being applied
 - f. include any conditions or building assessment provisions applicable to the building approval
 - g. include any restriction applying to the use of the building
 - h. if the building uses a performance solution, list the solution and the performance requirements of the NCC applicable including any required measures applying to the use of the building
 - i. if the building has a dispensation, outline the policy non-compliance or non-conformance including any required measures applying to the use of the building
 - j. list the referenced documents applicable to the building approval including the name of the person that prepared the document and the document identifier such as version number and date
 - k. include a preliminary schedule identifying all applicable fire safety measures including the applicable standard of performance i.e. NCC provisions, specific policy requirements as well as relevant standards, including all relevant year versions
 - l. identify the required inspections by the building surveyor and any relevant competent person, including a mandatory final inspection
 - m. state that the building approval expires 3 years from the date of the approval if the building work has not commenced
 - n. identify the building surveyor including accreditation number and level and relevant state or territory accreditation details and be signed and dated by that person.
- 3.38 The building surveyor shall provide the building approval and supporting evidence to the delivery authority or their delegated representative. The documents shall be stamped as approved or otherwise endorsed by the building surveyor. The building surveyor shall retain one set of the approved documents and provide an electronic copy or as otherwise specified to the delivery authority or their delegated representative. This documentation must be retained for a minimum period of 10 years from the issue date of the certificate of completion.
- 3.39 The delivery authority or delegated representative is to ensure a minimum of one set of approved documents is maintained on site for the period of building work. In addition, the

delivery authority shall save the building approval and supporting evidence in Objective prior to handover. This requirement does not limit further distribution of approved documents.

- 3.40 Other than exempt building work, no building work shall commence until a building approval has been issued by the surveyor.

Variation to a building approval

- 3.41 Where there is a proposed variation to the building work which would be inconsistent with the building approval, a variation to the building approval shall be made. In this instance the NCC version applicable in the original approval shall be adopted.
- 3.42 Any variation to the building approval shall be supported by the processes in this chapter including clearly marked documentation showing the variation.
- 3.43 If the proposed variation involves a performance solution or dispensation, the processes required in section 2 shall be followed.

Required inspections

- 3.44 The building surveyor shall determine the required inspections, including for demolition work, and shall list the required inspections in the building approval.
- 3.45 The required inspections of building work must include, but are not limited to the following measures required by the building approval:
- a. any stage or element required under the relevant state/territory building approval legislation
 - b. any inspections requested by any person other than the building surveyor who will be expected to certify that any part of the building work complies with the building approval including persons who prepared a performance solution report or dispensation report regarding:
 - i. structural stages/milestones
 - ii. fire/smoke separation construction
 - iii. sound rated construction
 - iv. witness testing of fire safety systems
 - v. witness testing of emergency evacuation lighting systems
 - vi. any other stages/inspections nominated within the building approval and final inspection.
- 3.46 The person in charge of the building work must arrange for an inspection when the relevant part is completed and ensure that it is able to be independently inspected for compliance and conformance.
- 3.47 An installation certificate must be prepared following all inspections required in the building approval. The installation certificate must be in accordance with the template. Installation certificates are to be provided to the policy contact on request.
- 3.48 Required inspections must be carried out by the building surveyor or a competent person. The final inspection must be carried out by the building surveyor.

- 3.49 Where a required inspection is undertaken and the work is identified as non-compliant the following process must be followed:
- a. the person inspecting the work must issue a written direction to the person in charge of the building work setting out the areas of non-compliance/non-conformance and either:
 - i. set out the work or actions required to bring the building work into compliance and conformance (required work)
 - ii. the documents required to support a variation to the building approval so building work will comply with varied building approval (required documentation).
 - b. The written direction must state:
 - i. that once any required work is carried out, a reinspection must be completed
 - ii. a specified time within which the required work must be carried out or the required documentation must be provided, no less than 60 days.
 - c. Where a reinspection is required, it must be attended by the building surveyor or a competent person.
 - d. If:
 - i. a reinspection is required, and the work has not been completed to the satisfaction of the competent person or the building surveyor; or
 - ii. documents required to support a variation to the building approval are not approved by the building surveyor or, where a performance solution or dispensation is proposed, by Defence

the building surveyor must notify the delivery authority or delegated representative and/or provide them with a proposal for how the work will be brought into compliance or conformance.
- 3.50 Where any work or action is required as part of the above process, the building surveyor must continue to comply with the conflict-of-interest requirements above.

Occupation during building works

- 3.51 Where building work is being undertaken in an existing building it should not be occupied by any person not involved in carrying out of building work unless there has been an occupation assessment to determine whether it is safe and suitable. The assessment may be completed by the building surveyor or another competent person e.g. a structural engineer and/or fire safety engineer. The assessment shall be in the form of a 'Construction Zone Safety Strategy' and shall include but not be limited to consideration of the following elements:
- a. structural integrity
 - b. fire/smoke barriers
 - c. access from occupied areas to exits
 - d. access from occupied areas to services and equipment e.g. hydrants, hose reels and portable fire extinguishers
 - e. isolation of fire safety systems such as sprinklers, smoke detection or smoke hazard management systems (staged or permanent) within the area of building work or in the occupied areas

- f. access to amenities
 - g. a strategy to deal with issues identified in the assessment of items a. to f.
- 3.52 Issues such as compromised structural integrity or fire/smoke barriers, blocked exits, access to services or isolation of fire safety systems will need a strategy developed and measures agreed and implemented to ensure the building remains safe for continued occupation.
- 3.53 Support to continue or commence occupation based on the strategy developed shall be sought from the project stakeholders and the authority responsible for the personnel who occupy the building.

Certificate of completion

- 3.54 A certificate of completion may only be issued for building work that has received a building approval. A separate certificate of completion is required for each individual building.
- 3.55 A certificate of completion can be issued by the building surveyor when:
- a. the required inspections have been undertaken, and the building surveyor is satisfied that the building work is complete
 - b. supporting evidence of installation/construction compliance and conformance from designers, installers and builder, including installation certificates required by the building approval, has been received and determined to be acceptable by the building surveyor
 - c. verification from the person responsible for the performance solution or dispensation report that any requirements are installed and implemented has been received and determined to be acceptable by the building surveyor
 - d. reasonable evidence is provided that any requirement or condition included within the building approval has been complied with, or that requirement or condition has ceased to apply.
- 3.56 A certificate of completion must:
- a. identify the building location and property details including the GEMS identification
 - b. reference the applicable building approval and any variations to the building approval
 - c. include description of building work NCC occupancy classifications:
 - i. type of construction
 - ii. large-isolated building status
 - iii. floor area
 - iv. rise in storeys
 - v. effective height
 - vi. contribution factor
 - vii. identify the aircraft information provided on the Hangar Assessment Form for an aircraft hangar, if applicable
 - d. include any restriction applying to the use of the building
 - e. environmental requirements, if applicable

- f. if the building uses a performance solution, list solution and performance requirements applicable including any required measures applying to the use of the building
 - g. if the building has a dispensation, outline the policy non-compliance or non-conformance including any required measures applying to the use of the building
 - h. identify the building surveyor including accreditation number and level and relevant state or territory accreditation details and be signed and dated by the building surveyor
 - i. include a schedule identifying all applicable fire safety measures which must identify the applicable standard of performance i.e. NCC provisions, specific policy requirements as well as relevant standards, including all relevant year versions.
- 3.57 The building surveyor shall provide an electronic copy of the certificate of completion and supporting evidence or as otherwise specified to the delivery authority or their delegated representative for retention on the project file. The delivery authority or their delegated representative is to ensure that the staged, interim or final certificate of completion and supporting evidence.
- 3.58 The building shall not be occupied or used until a certificate of completion has been issued which applies to the part of the building to be occupied or the entire building.
- 3.59 The building shall not be occupied or used in a manner that is inconsistent with the certificate of completion.
- 3.60 A copy of the certificate of completion and amendments as a result of new building work shall be provided and displayed as near as practicable to the building's main entry.

Staged or interim certificate of completion

- 3.61 Where permitted under relevant state or territory legislation, a staged or interim certificate of completion shall comply with the requirements of clauses 3.57 and 3.58 and may be issued where a building is to be completed and occupied in stages. The part of the building occupied must comply with the applicable building approval requirements and have a staged or interim certificate of completion. The safety of occupants of that part must not be compromised by future works or vice versa.
- 3.62 The staged or interim certificate of completion will expire on the earliest occurrence of any of the following events:
- a. written cancellation by the building surveyor
 - b. issue of the final certificate of completion
 - c. the date nominated on the staged or interim certificate of completion.
- 3.63 If a staged or interim certificate of completion expires, occupation of the building (or part) is no longer permitted unless the process in clause 4.62 is repeated.
- 3.64 Where staged or interim certificates of completion are not permitted under relevant state or territory legislation, a separate building approval must be issued for each stage of the work.

Exempt building work

- 3.65 The building work listed is exempt unless otherwise required by the applicable state/territory.
- 3.66 While exempt building work is not required to comply with this manual, all work must still be completed in accordance with the applicable design standards and codes, for example:

- a. attaching sunscreens or sunhoods
 - b. attaching a sunscreen or sunhood to an existing building is exempt if the sunscreen or sun hood area is less than 2m²
 - c. particular work for class 10 buildings or structures.
- 3.67 An exemption applies to the building work of above-ground rainwater tanks less than 3 metres in height providing a non-potable water supply.
- 3.68 Buildings having a floor area not exceeding 10m², except those that are defined as CF1 or CF2, are EO storage and/or sited in a “cyclonic zone/area”, are exempt. Cyclonic areas are those determined as being in wind regions C and D in accordance with AS/NZS 1170.2.
- 3.69 Building work for a class 10b structure or special structure is exempted if:
- a. the structure is sporting or playground equipment
 - b. physical training equipment
 - c. minor plant and equipment covers
 - d. the structure is no higher than 3 metres above its natural ground surface.

Repairs, maintenance or alterations not affecting structural component or fire safety system.

- 3.70 Building work that consists of repairs, maintenance or alterations to an existing building, is exempt if they are not a substantial alteration and they do not:
- a. change the building’s floor area or height
 - b. alter egress paths or evacuation processes
 - c. change the building classification under the NCC
 - d. affect fire safety provisions required under the NCC or local regulations
 - e. affect structural components
 - f. affect fire safety systems, required by a Defence policy, of the building.
- 3.71 Exception examples:
- a. an internal refurbishment/changing office/fixed equipment
 - b. an installation/alteration of an air-conditioning system that does not affect any fire safety system.

Repairs, maintenance or alterations only affecting a minor component of a fire safety system.

- 3.72 Building work that consists of repairs, maintenance or alterations to an existing building are exempt if they:
- a. comply with the provisions above, excluding paragraphs d. and f.
 - b. only affect a minor component of the building’s fire safety system.
- 3.73 A minor component means an element of the fire safety system that is required for the system and is not an addition.
- 3.74 Any changes to the location of fire safety system components must be documented and form part of baseline data for ongoing maintenance requirements.

- 3.75 Examples of a minor component of a fire safety system include, but are not limited to, changes to a single sprinkler head or a smoke detector or single elements of other required safety systems listed on the buildings fire safety measures schedule.

Deployable infrastructure

- 3.76 Tents, membrane structures, deployable hangars or other deployable infrastructure are exempt if:
- a. the floor area is less than 100m² (no time limit applies)
 - b. erected and removed within 2 months (no area limit applies)
 - c. they are only used for a military training or operational activity.

Work for particular non-loadbearing devices

- 3.77 This exemption applies to building work for the following devices if they are non-loadbearing and outside restricted areas, including but limited to hazardous object zones required for aircraft:
- a. an aerial or antenna
 - b. a satellite dish with a maximum diameter of 900mm
 - c. a flagpole, mast or tower.
- 3.78 The exemption to these devices only applies if the device is:
- a. attached to a building and no more than 3.5 metres above the highest point of fixing building
 - b. detached from any building and no more than 10 metres above the device's finished ground surface.

Work for particular retaining walls

- 3.79 Building work for a retaining wall is exempt if:
- a. there is no surcharge loading over the zone of influence for the wall
 - b. the total height of the wall and of the fill or cut retained by the wall is no more than 1m above the wall's finished ground surface
 - c. the wall is no closer than 1.5 metres to a building, swimming pool fence or another retaining wall.
- 3.80 A surcharge loading means a load applied to a soil stratum that has, or may have, the effect of consolidating the stratum, other than a load arising only from:
- a. persons or vehicles on, or moving over, the stratum
 - b. the effects of rain on the stratum.
- 3.81 A zone of influence, for the retaining wall, means the volume of soil stratum behind the wall that affects the wall's structural integrity.

Work for particular signs

- 3.82 Building work is exempt if it consists of the erection of a sign that is detached from a building and is no higher than 2 metres and no wider than 1.2 metres.

- 3.83 Work for particular temporary structures on building sites is exempt if it is for:
- a. a temporary site office or amenities provided for the duration of building work on or serving a building site
 - b. a transportable building that has a maximum floor area not exceeding 100m² for a maximum period of 6 months
 - c. a gantry
 - d. scaffolding.

Work for particular fences

- 3.84 Construction of a fence is exempt if the fence is not for a swimming pool.

Non-Defence estate buildings

- 3.85 The following paragraphs sets out the requirements for Defence-occupied buildings that are not on the Defence estate. These include requirements for:
- a. pre-lease inspection and reporting
 - b. building work for Defence
 - c. maintenance requirements.

Pre-lease inspection and reporting

- 3.86 All new leases and leases subject to renewal negotiation shall be assessed against the requirements of the current IEM and the construction code that applied at the time of building approval. Compliance and conformance shall be confirmed prior to entering into the new lease or lease renewal.
- 3.87 A certificate of completion and any performance solutions shall be obtained by the Property Management Branch or its delegated representative to ensure the building has been certified as fit for use.
- 3.88 A full NCC audit which includes an IEM fire safety survey report is to be undertaken to confirm suitability prior to entering a lease or lease renewal. The audit is to:
- a. be undertaken by a building surveyor
 - b. have regard to the certificate of completion and any performance solutions
 - c. identify any combustible cladding on a building of type A or type B construction.
- 3.89 If the Building Contribution Factor Form is not available prior to the lease, the capability manager is required to provide an indication of the contribution factor for the building as input into the NCC/IEM audit.
- 3.90 Where combustible cladding is identified in a building of type A or type B construction, the following requirements must be met:
- a. replace all combustible cladding as necessary to achieve compliance with the deemed-to-satisfy provisions of the NCC
 - b. where retention of any combustible cladding is proposed, then comply with the processes identified in 'Guidelines for external wall combustible cladding' located at the Building Works Section page.

Note: Evidence that remediation works will or have been completed and certified in compliance with state/territory building legislation must be submitted to Defence for review.

- 3.91 The Assistant Secretary Property Management Branch and/or Assistant Secretary Regional Services are responsible for ensuring that the above process has been completed.

New building work

- 3.92 All building work relating to Defence occupied buildings shall comply with relevant state, territory or other authority building legislative processes.
- 3.93 Where building work is proposed within a Defence occupied building, Defence must be a stakeholder in the building approval process for any proposed performance solutions or dispensations to Defence policy requirements set out in Chapter 2.

No building work

- 3.94 Defence occupied buildings that are not the subject of any building work must continue to comply with any certificates of completion.
- 3.95 The intention is not to apply IEM design requirements to existing buildings unless they are the subject of building work.

Maintenance

- 3.96 Defence occupied buildings must be maintained in accordance with the requirements of any certificates of completion for the building. Evidence that this maintenance is or will be undertaken must be provided to Defence at least annually.
- 3.97 Where fire safety measures are installed as a result of NCC or the Defence compliance and conformance requirements outlined in this chapter, those services must be maintained in line with this IEM Manual relevant guidelines and with any associated certificates of completion for the building. This applies to:
- a. aircraft hangers
 - b. special structures
 - c. certificate of regularisation

Defence specific requirements

- 3.98 Guideline 3000 – Certification of Building Works and Guideline 3001 – External wall cladding articulate Defence specific requirements relevant to this chapter.

SUSTAINABILITY RATINGS SCHEMES

Background and Context

- 3.99 Administered by the Department of Finance, the [Net Zero in Government Operations Strategy](#) requires certain Commonwealth-owned and leased buildings to have an appropriate energy performance rating from a sustainability ratings scheme provider.
- 3.100 Not all buildings are subject to the above, with their size and usage key in informing participation, or the nature of participation, in a rating scheme.
- 3.101 NABERS, Green Star and Infrastructure Sustainability Council provide rating tools that can be a means for Commonwealth entities to understand and reduce property-related emissions to enhance sustainability outcomes.
- 3.102 The rating tools gauge an organisation's performance in response to reducing emissions. They aim to achieve alignment with established sustainability principles through design, implementation and operational phases of any built form.
- 3.103 Defence is committed to aligning with the Net Zero in Government Operations Strategy, where practical, and where it does not impact operational capability and requirements.
- 3.104 For some buildings or facilities, participation in a rating scheme may pose a risk to Defence, including physical and data security, and the provision of information, such as layout and performance metrics to third parties.
- 3.105 In this case, and as is available to Defence through the Net Zero in Government Operations framework, Defence may apply an alternative approach to deliver the same sustainability outcomes.

Policy Statement

- 3.106 Defence will deliver sustainability by applying or exceeding the mandatory NCC on buildings and facilities for:
- a. construction
 - b. leasing
 - c. refurbishment
 - d. data centres.
- 3.107 As an overarching principle, Defence supports the intent of a rating scheme to measure performance against net zero objectives and will participate provided it does not:
- a. conflict with the Defence Security Principles framework; and/or
 - b. have the potential to impact Defence's operational or capability requirements.
- 3.108 Where it is not appropriate to participate in a rating scheme, Defence will apply the Defence Built Environment Sustainability Approach (DBESA). This approach supports Defence to meet its sustainability and net zero goals, delivering outcomes consistent with Government policy.

Infrastructure Construction and Refurbishment

- 3.109 A Defence building or facility may participate in a rating scheme provided:
- a. the building or facility is classed at Security Level Zone 2 or below; and

- b. the participation of a building or facility in a rating scheme will not impact operational or capability requirements.
- 3.110 The DBESA will apply to buildings or facilities classed as Security Level Zone 3 or above.
- 3.111 Approval for participation in a rating scheme or the DBESA, for a Defence construction or refurbished building or facility, must be obtained from the ASEE.

Property Leasing

- 3.112 Defence may participate in rating schemes for:
 - a. new leases
 - b. renewals
 - c. sub-leases.
- 3.113 Defence will support leases (with Green Lease Schedules) that include rating schemes for a building or facility where:
 - a. the building or facility is classed as a Security Level Zone 2 or below; and
 - b. the participation of a building or facility in a rating scheme will not impact operational or capability requirements.
- 3.114 In a building or facility with mixed security zones, where Defence is the sole occupant, the highest zone classification is considered to apply to the entire building or facility for the rating scheme.
- 3.115 In a building or facility where Defence shares occupancy with other non-Defence tenants, Defence will support leases that include rating schemes if:
 - a. the Defence tenancy is classed Security Level Zone 2 or below; and
 - b. the participation of a building or facility in a rating scheme will not impact Defence operational or capability requirements.
- 3.116 For leased buildings or facilities classed at Security Level Zone 3 and above, Defence will apply the DBESA. Green Lease Schedules will not apply. For leased buildings or facilities negotiated on behalf of portfolio agencies, Defence will apply the DBESA.
- 3.117 Approval for participation in a rating scheme or the DBESA for a Defence leased building or facility must be obtained from ASEE.

Data Centres

- 3.118 The DBESA will apply to all Defence owned, operated and or leased data centres. Approval for the DBESA must be obtained from ASEE.

Guidance and Support

- 3.119 Queries about Sustainability in Defence should be directed to:
sustainability.defence@defence.gov.au.
- 3.120 Additional guidance materials to support the implementation of this policy are set out in the Guideline 3002 - Sustainability Rating Guidance.

CHAPTER 4

MECHANICAL SERVICES

GENERAL PRINCIPLE

- 4.1 This chapter specifies the mechanical services compliance and conformance requirements for design, construction and maintenance of Defence occupied buildings for the protection of personnel and assets in support of Australian Defence objectives.

Rationale

- 4.2 The rationale of this chapter is to:
- a. reinforce the requirement that building work on the Defence estate and other Defence occupied buildings must comply with the NCC and all relevant Defence policies
 - b. align the administrative requirements for building work on the Defence estate to state and territory building legislation and certification requirements as much as practicable.

Expected outcome

- 4.3 Mechanical systems shall comply with all relevant state legislation, the NCC, and applicable Defence policies and standards governing the design, construction, modification, and maintenance of building services and infrastructure.

Cooling and heating capacity requirements

- 4.4 As a minimum the following allowances should be applied:
- a. data centres: N+1 cooling
 - b. communications rooms: Supporting CF1-2 facilities N+1 cooling, supporting CF3-5 facilities N cooling
 - c. central cooling plant and heating plant: minimum 2 items of equipment with 60% capacity of the peak diversified load
 - d. systems with light loads: plant with demonstrated high turn down for light load cooling or heating.

Spare capacity

- 4.5 For HVAC systems, a minimum of 5% spare latent and sensible capacity shall be provided to heating and cooling load calculations:
- 4.6 For ventilation systems a minimum of 5% spare airflow shall be available for additional equipment via system adjustment.

Equipment lifespan

- 4.7 The benchmark equipment life spans for Defence buildings are:
- a. minimum 20 years for packaged dx plant
 - b. minimum 25 years for chilled water plant

- c. minimum 15 years for VRF plant and split ducted plant
- d. minimum 10 years for split dx plant.

Electric duct heaters

- 4.8 The use of electric duct heaters is to be minimised to support Australian Government energy minimisation and emissions targets.

HVAC SELECTION AND DESIGN CONSIDERATIONS

Offices

- 4.9 Defence offices should incorporate:
- a. ducted variable air volume systems
 - b. VRF or chilled water systems (chilled water can be provided for smaller offices and must be provided for offices greater than 1000m² of air-conditioned space)
 - c. separate air handling plant for perimeter and internal zones
 - d. air handling plant located in plant rooms
 - e. general design to meet Property Council of Australia A grade allowances and the NAMP/DOAG
 - f. condensing units and chillers shall be in screened secure plant enclosures
 - g. mould minimisation methods subject to Defence approval.

Living In Accommodation (LIA)

- 4.10 Air conditioning systems for LIA shall comprise individual, separately controlled systems for each bedroom and common room. In each LIA bedroom, an occupant activated device such as a key switch shall be provided to operate the air conditioning unit and other HVAC equipment. This shall comply with the requirements of NCC (BCA) Section J.
- 4.11 Defence classifies LIA bedrooms as Category I Occupancy as defined under ISO 817 and AS/NZS 5149 series Categories of Occupancy. As such, for group A1 and A2 refrigerants there shall be a restriction of refrigerant charge.

Category classification of communications rooms and similar IT Rooms

- 4.12 All communications and IT rooms are to be considered normally occupied spaces in relation to HVAC design and refrigerant concentration.

Maintenance access

- 4.13 All piped services and valves are to be accessible for routine inspection and maintenance, and for replacement at end-of-life without building modification. Piping shall not be installed in shafts without access for removal.

Cooling water systems

- 4.14 Cooling water systems are to be installed and maintained as per the AS/NZS 3666.

HVAC outdoor design conditions

- 4.15 Selection of the most appropriate outdoor design conditions is critical to the correct sizing of an HVAC plant. Defence requires that HVAC plants be correctly sized to provide fit for purpose design solutions that provide the most economical solution on a WOL basis. There are 2 scenarios to consider:
- a. Defence facility location included in AIRAH DA9: Outdoor design conditions can be based on the values presented in AIRAH Application Manual DA9 Air Conditioning Load Estimation and Psychometrics (AIRAH DA9), if this publication provides design data for the site of the proposed Defence facility
 - b. Defence facility location not included in AIRAH DA9: Where AIRAH DA9 design conditions are not scheduled for the Defence building location the service provider needs to determine and justify the proposed outdoor design conditions. If the Defence facility location is not included in AIRAH DA9 the service provider can determine outdoor design conditions based on:
 - i. the nearest location in AIRAH DA9 to the Defence facility for which outdoor design conditions are available
 - ii. the nearest location to the Defence facility for which Bureau of Meteorology (BoM) weather data is available for 8.00am and 3.00pm conditions.

Note: If appropriate weather data is available for a BoM location closer to the Defence facility than the nearest location in AIRAH DA9 and in a similar locality (e.g. altitude, distance from sea) then data for this BoM location can be used.

- 4.16 Some Defence buildings and projects (e.g. critical facilities sensitive to outdoor temperature increase and with long-term project lives) will require outdoor design conditions for air conditioning calculations to be adjusted to allow for probable effects of climate change.

Note: Current information indicates that an appropriate allowance for the probable effects of climate change by 2030 is for the summer dry bulb and wet bulb temperatures used in current air conditioning cooling load calculations to be increased by between +0.5°C and +2°C.

CHAPTER 5

FIRE ENGINEERING

GENERAL PRINCIPLE

- 5.1 This chapter describes the Defence standards and procedures that enable the application of fire safety engineering policy. Requirements are to be applied retrospectively where required.
- 5.2 Achieving adequate levels of fire safety across the Defence estate is technically complex and requires the involvement and oversight of skilled competent practitioners, including but not limited to fire safety and building surveying professionals. This chapter specifies the minimum Defence fire safety requirements in support of Defence objectives, namely enabling and supporting the sustainment of capability via the provision, maintenance and renewal of installations, systems and infrastructure that are regulatory compliant, safe, functional and fit for purpose, sustainable, energy efficient and offering the optimum through-life performance.

Rationale

- 5.3 The rationale of this chapter is to:
- a. reinforce the requirement that building work on the Defence estate and other Defence-occupied buildings must comply with the NCC and all relevant Defence policies
 - b. align the administrative requirements for building work on the Defence estate to state and territory building legislation and certification requirements as much as practicable.

Expected outcomes

- 5.4 Fire systems shall comply with all relevant state legislation, the NCC, and applicable Defence policies and standards governing the design, construction, modification, and maintenance of building services and infrastructure.
- 5.5 In addition, the fire safety measures outlined in this section will be implemented to protect personnel, assets and areas critical to the operation of military platforms in support of Australian Defence objectives.

General requirements

- 5.6 This chapter outlines specific fire protection measures and fire safety policies that are determined based on life safety risk and building criticality. These measures may exceed the minimum requirements of the NCC. It also includes operational and maintenance requirements, as well as guidelines for the disposal of buildings. Additional asset protection requirements apply to buildings classified as CF1 or CF2, as detailed within this manual.

Fire safety strategy

- 5.7 Any CF1 or CF2 building or part that is not covered in this manual or associated guidelines shall have a fire safety strategy developed to protect it from a fire event. The proposed strategy must be agreed by the DEEP prior to a building approval being issued.

Fire Safety Management Strategy (FSMS)

- 5.8 Where non-compliances are identified through the Fire Safety Survey (FSS) program rectification through building works may not always be feasible or viable for Defence. In such

cases, the implementation of the FSMS may be adopted as a temporary solution. Further guidance is provided in [Guideline 5001](#).

Reporting of Fire

- 5.9 The Commonwealth *Work Health and Safety Act 2011* (Part 3) defines an uncontrolled fire as a dangerous incident that must be reported to Comcare. Guidance on responding to and reporting incidents is available at the Comcare web page under responding to an incident. Uncontrolled fires must also be reported to Defence using Sentinel.

Certification and verification of works

- 5.10 The certification and verification of all works, designs, construction, commissioning, operations, maintenance, and decommissioning must be in accordance with all statutory requirements and Defence policy requirements. The delivery agent must be responsible at all times of ensuring this is complied with and in providing the necessary assurances and confirmation as and when requested.

SMOKE DETECTION AND ALARM SYSTEM

General requirements

- 5.11 Fire detection and alarm systems must be installed where required to meet the performance requirements of the NCC. The smoke detection and alarm systems must be provided to facilitate early warning and enhance asset protection in all CF1 and CF2 buildings or any building containing CF1 or CF2 parts.
- 5.12 Defence-specific requirements for fire detection and alarm systems are detailed for certain building types, including:
- a. computing and electronic equipment, CF1 and CF2 ([Guideline 5003](#))
 - b. storage and workshop buildings, CF1 and CF2 ([Guideline 5004](#))
 - c. aircraft hangars ([Guideline 5005](#))
 - d. building construction in bushfire-prone area ([Guideline 5006](#))
 - e. explosive ordnance buildings ([Guideline 5007](#))
 - f. Defence living-in accommodation ([Guideline 5008](#))
 - g. enclosed carparks of electrical vehicle and lithium-ion storage battery facilities ([Guideline 5010](#))
 - h. special structures ([Guideline 5012](#)).
- 5.13 Fire detection systems must integrate with building occupant warning systems (BOWS) or emergency warning and intercommunication systems (EWIS) in accordance with the above relevant standards.
- 5.14 Fire detection and alarm systems shall be considered in other facilities on an individual basis when such an addition would provide an asset protection benefit and/or improved safety at reasonable cost. The consideration must be documented in the project design report.

Monitoring Requirements

- 5.15 All fire suppression and detection systems must be connected to a monitoring service provider, fire station dispatch centre or an appropriately staffed monitoring point where

available. Monitoring is not required for remote facilities that are not provided with a fire service response or any staff to monitor alarms in a building.

- 5.16 Where Defence base fire brigades or first responders' facilities are available, all fire alarms are to terminate at the fire alarm console of the fire station. Where this occurs, the fire alarm monitoring system must comply with relevant Australian Standard.
- 5.17 Where no permanently staffed Defence base fire brigades or first responders' facilities are available, alarms are to terminate at a fire alarm console in an appropriately staffed monitoring point on the base or installation where available.
- 5.18 Connection from the console to the public authority fire service to provide automatic alerting is not necessarily required. Where possible, mutual aid arrangements should be developed so that support may be obtained from the public authority fire service where available.

Note: Each facility does not require a separate ASE connected to the building's fire detection control and indicating equipment (FDCIE). The designers may consider the options to interface the sub-FDCIEs with the main FDCIE by other means as required in the Australian Standards.

Non-required smoke detection system

- 5.19 Where a smoke detection and alarm system is installed and is not a requirement of this manual or the NCC, the system may be considered for decommissioning and removal. Further guidance is provide in [Guideline 5011](#).

Note: Decommissioning a fire protection system is not considered as work that is exempt from a Building Approval.

BUILDING EMERGENCY WARNING SYSTEMS

Defined terms

- 5.20 For the purposes of this section, the term "emergency warning system" encompasses:
 - a. a BOWS, which is mandated in conjunction with fire detection and/or sprinkler systems
 - b. an EWIS.

General requirements

- 5.21 A BOWS must be installed in Defence buildings where building works or change-in-use occurs, as required by the NCC. It is also required where this manual or associated guidelines specify sprinkler system or smoke detection systems over and above the requirements of the NCC.
- 5.22 All new BOWS must be installed in accordance with the NCC relevant Australian design standards, and must include:
 - a. a public address (PA) system with a service microphone that is audible throughout the building. In particular:
 - i. the PA module must be located adjacent to or within the FDCIE
 - ii. the PA function must comply with the requirements of relevant design standards for Grade 3 equipment
 - iii. the PA function must be secured from unauthorised persons by a locked door, with the key being identical to other keys required for access to the FDCIE
 - iv. the system must be classified as an essential service.

- 5.23 The PA system connected to a BOWS is not required to have intercom systems. Intercom systems are only required for EWIS.
- 5.24 Visual warning devices or warning signals for the hearing impaired, provided by means other than loudspeakers, as appropriate.

Requirements with sprinkler or smoke detection systems

- 5.25 Where smoke detection and/or sprinkler systems are provided, a BOWS is also required. Refer to clauses S17C8 and S20C7 of the NCC.

Emergency Warning and Intercom System (EWIS)

- 5.26 Several NCC building classifications are mandated to install a EWIS over BOWS.
- 5.27 Where required by NCC, all EWIS must be installed in accordance with the NCC and relevant Australian standard and must include:
- a. red fire alarm manual call points, which must be connected to both the fire detection and emergency warning systems. The use of white emergency call points, which are connected only to the warning system, must not be used
 - b. visual warning devices or warning signals for the hearing impaired, provided by means other than loudspeakers, as appropriate.

Existing buildings

- 5.28 The installation of a BOWS is also required in existing Defence buildings that are equipped with a fire detection system and/or a sprinkler system, where an internal warning system is not already provided. The system must meet the applicable provisions of the NCC and the relevant Australian standards.

FIRE HYDRANT AND HOSE REEL SYSTEMS

General requirements of fire hydrants & fire hose reels

- 5.29 Fire hydrant systems and fire hose reels shall be installed in Defence buildings in accordance with the relevant standards.

Emergency vehicular access and hydrant system requirements

- 5.30 Emergency vehicular access and hydrant systems are to comply with the requirements of the NCC and related Australian Standards in force at the time of building approval and the operational requirements of the responding fire service. The delivery authority for new projects must ensure the requirements of all responding fire services for vehicular access, including pavement load bearing capacity, and hydrant systems are met. The evidence of consultation with responding fire services must be documented in the project design report.

PORTABLE FIRE EXTINGUISHERS (PFES) AND FIRE BLANKETS

General requirements of PFES

- 5.31 PFES and blankets are to be strategically located where they are most needed to support first-attack firefighting efforts and limit the consequences of fire on property, personnel, and operations.
- 5.32 PFES must be distributed throughout Defence buildings in accordance with the relevant NCC provisions and specific requirements for various fire risks and building classifications.
- 5.33 The distribution of PFES must comply with the following:

- a. Class 1b buildings: PFEs must be distributed as if the building were classified as Class 2–9 buildings, following the distribution requirements outlined in E1D14 of the NCC.
- b. Dangerous good/Hazardous chemicals: PFEs must be installed in areas where dangerous goods or hazardous materials are stored, handled, or used in accordance with the provisions of the relevant standard
- c. Fume cupboards: where fume cupboards are present, PFEs must be provided in line with the guidelines specified in AS/NZS 2243.8.

Location requirements

- 5.34 The installation and location of PFEs is critical for ensuring they are readily accessible in the event of a fire emergency. The following guidelines must be followed to ensure compliant distribution within Defence buildings:
- a. Where only one PFE is required, it should be installed on an inside wall, adjacent to the normal access/egress door to the installation. This ensures that the PFE is readily accessible upon entry
 - b. Where more than one PFE is required, they should be located in the following order of preference:
 - i. on an inside wall, adjacent to the normal access/egress door
 - ii. on an inside wall adjacent to alternative access/egress doors
 - iii. on inside walls that maintain the required travel distance as defined by the NCC to ensure quick and safe access
 - iv. on the wall outside the main entry door, where they should be suitably protected from weather, theft, and vandalism to ensure readiness and availability.

Installation requirements

- 5.35 The designers must follow these requirements in addition to NCC requirements:
- a. PFEs must be securely mounted on walls or on other appropriate surfaces to ensure stability and prevent accidental dislodging
 - b. PFEs must be clearly visible and accessible, with signs and markings indicating their location
 - c. installation height for PFEs should be in accordance with the NCC, ensuring that they are easily accessible to personnel of varying physical abilities
 - d. PFEs must provide the location as well as type signs.

General requirements for fire blankets

- 5.36 Fire blankets must be installed in accordance with the location requirements specified in the relevant standards, which includes placing them in easily accessible areas, close to the source of the fire risk (e.g. near cooking appliances). They must be clearly marked with appropriate signage and instructions for use.

Note: Fire blankets are not required within sole-occupancy units of Class 2 or Class 3 buildings, nor are they required in Class 4 sole-occupancy unit parts.

FIRE SUPPRESSION SYSTEMS

General requirements

- 5.37 All fire suppression systems must conform to the NCC and relevant Australian Standards. The compliance with Defence-specific requirements, as detailed in this manual, is mandatory.
- 5.38 Where there is deviation from the NCC deemed-to-satisfy provisions performance solutions must be developed to address specific operational needs.
- 5.39 Defence-specific requirements for fire suppression systems are detailed for certain building types, including:
- a. computing and electronic equipment, CF1 and CF2 ([Guideline 5003](#))
 - b. storage and workshop buildings, CF1 and CF2 ([Guideline 5004](#))
 - c. aircraft hangars ([Guideline 5005](#))
 - d. building construction in bushfire-prone area ([Guideline 5006](#))
 - e. explosive ordnance buildings ([Guideline 5007](#))
 - f. Defence living-in accommodation ([Guideline 5008](#))
 - g. enclosed carpark of electrical vehicles and lithium-ion storage battery facilities ([Guideline 5010](#))
 - h. special structures ([Guideline 5012](#)).

Note. Firefighting foams containing PFAS shall not be used in new installations.

- 5.40 The fire suppression systems it to be monitored through a fire station dispatch centre or an appropriately staffed monitoring point to facilitate prompt emergency responses.

Design requirements

- 5.41 The fire suppression systems must incorporate environmentally responsible containment and disposal mechanisms to prevent contamination. In aircraft hangars and fuel storage areas, floor drainage systems must be equipped with flame traps, while hydrocarbon detection systems must be integrated into drainage infrastructure where required.
- 5.42 The designers must consider the environment when proposing the options for the fire suppression systems for the projects.
- 5.43 The fire suppression systems must be interfaced with fire detection and alarm systems, or any other systems where required by the design's intent.
- 5.44 The fire suppression systems must be integrated with the BOWS or EWIS in accordance with the above referenced standards.
- 5.45 All fire suppression systems must be commissioned in accordance with the relevant performance solution requirements, referenced design and maintenance standards.
- 5.46 The designers in early design stages and during the commissioning process must verify water supply adequacy and redundancy, detection and activation mechanisms, and control station functionality.

EGRESS AND PHYSICAL SECURITY REQUIREMENTS

General requirements

- 5.47 This section applies to:
- a. doorways in a required exit as defined in the NCC
 - b. doorways forming part of a required exit
 - c. doorways or security barriers in the path of travel to a required exit
 - d. egress through barriers in the path of travel to open space.
- 5.48 Exit doors that need to be maintained includes required doors that provides access to, or is within, a fire-isolated stairway or ramp; or a fire-isolated passageway; or a required doorway which leads to road or open space.
- 5.49 A doorway forming part of a required exit means a doorway or door that provides access to, or is within, a fire-isolated stairway or ramp; or a fire-isolated passageway.
- 5.50 A doorway or door in a path of travel to an exit is any door that a building occupant must pass through to reach the exit from the storey with the exception of those doors permitted to be excluded under D3D26(3) of the NCC.
- 5.51 Open space is as defined in the NCC.

Means of egress

- 5.52 Means of egress from Defence buildings must comply with the requirements of the NCC.
- 5.53 Required levels of security may be achieved through:
- a. mechanical locks
 - b. electromechanical locks
 - c. electromagnetic locks in combination with mechanical locks.

Mechanical Locks

- 5.54 Mechanical locks can be installed on exit doors provided they comply with the NCC. Doors required by the NCC to be fitted with panic bars must not have locks that inhibit the operation of the panic bar. Mechanical locks must not restrict egress from the side of the door facing the person seeking to exit. The use of Redlam bolts or similar devices is prohibited on 'exits' and doors in the path of travel to an exit.

Requirements for doors in the Security Zones

- 5.55 In Government buildings, the Security Zones 4 or 5, hardware on exit doors, or doors in the path of travel to an exit, may use the concession of the NCC-particular clause provided that the door can be immediately unlocked:
- a. by operating a fail-safe control switch, not contained within a protective enclosure, to actuate a device to unlock the door
 - b. by a person or persons, specifically nominated by the owner, properly instructed, and available at all times when the building is lawfully occupied, to ensure that persons will be able to immediately escape if there is a fire.

Manual override devices

- 5.56 Where the option in clause 5.61 (a) is adopted, the manual override device must be capable of immediately releasing the lock via an action in a security control room or by a break glass unit (BGU) installed on the side of the door facing the person seeking egress.
- 5.57 BGUs must:
- a. be located between 0.9 metres and 1.1 metres above finished floor level and adjacent to the door hardware side of the door
 - b. be green in colour
 - c. release the locking mechanism of the door
 - d. include battery back-up to ensure the lock releases in the event of a power failure
 - e. include signage stating "Emergency Door Release" or similar, with white lettering on a green background.

Note: Exits may be wired to a security system and sounders, if required.

Fail safe operation

- 5.58 Security latches on doors in areas that are not classified as Australian Government Security Zones 4 or 5 must release in fire mode. The fail-safe operation must be activated by the building smoke detection and alarm system installed and/or the building sprinkler system.

Egress to public road

- 5.59 Where Defence buildings have secure perimeter fencing or similar, a means of egress to a public road must be provided in accordance with relevant clause of the NCC. The road infrastructure on a Defence site may be considered a public road.

Performance solutions

- 5.60 Where Defence's mandatory security requirements result in non-compliance with the egress related deemed-to-satisfy provisions of the NCC, a performance solution process, as outlined in this manual, must be considered.

Fire-resistant door-sets

- 5.61 Door hardware must not be installed on fire-resistant door-sets unless it complies with the egress requirements outlined in this section and has been tested. Testing must be done in accordance with the requirements of the above-mentioned standards, ensuring compliance with the prototype fire door-set.

MAINTENANCE

General requirements

- 5.62 Routine maintenance and service of the fire protection systems and equipment shall be carried out in accordance with the relevant standards unless otherwise approved in writing by the Director Estate Engineering Policy.
- 5.63 All the fire systems installed shall undergo the routine maintenance and service in accordance with the relevant standards. The maintenance requirements of any individual fire protection systems are not repeated in this clause.

- 5.64 Where any of the Defence-occupied buildings or part of a building are located on non-Commonwealth land, maintenance must also comply with state or territory regulations.

Minimum performance standards

- 5.65 All active and passive fire safety systems shall be maintained to meet their original design and operational requirements.
- 5.66 Any identified deficiencies shall be rectified promptly to ensure continued fire protection effectiveness. On completion of defect rectification, the system or component shall be tested for correct function.

Record keeping & reporting

- 5.67 Maintenance activities shall be documented and retained to comply with the routine services records requirements of the referenced standards as well as any contractual obligations.

Maintenance of the fire protection systems

Fire detection & alarm systems

- 5.68 Systems shall be maintained in accordance with the referenced standards to ensure reliability. Periodic functional testing, detector calibration, and fault condition analysis shall be carried out as required by the relevant standards.
- 5.69 If a non-required smoke detection system is identified as being at the end of its life, Defence prefers it is de-commissioned. For further guidance refer to [Guideline 5011](#).
- 5.70 Strategies to reduce false alarms should be reviewed and refined as part of routine servicing.

Fire doors & exit pathways

- 5.71 All required exits along paths of travel must undergo yearly inspections to verify that the door opens and closes freely and door hardware complies with the requirements of this section.
- 5.72 Testing includes, but is not limited to:
- a. a functional test of any electronic or mechanical hardware
 - b. the operations of any door latching hardware.

Solid-core doors

- 5.73 The solid core doors in living-in accommodation shall be inspected yearly to ensure the doors open and close freely. This includes, but is not limited to:
- a. a functional test of the mechanical hardware, self-closers
 - b. the operation of any door latching hardware.
- 5.74 The doors are to come to the fully closed and latched position after each manual opening. Any deficiencies must be reported and rectified.

Foam fire suppression systems in installed facilities

- 5.75 Foam suppression systems shall be tested and maintained in accordance with the referenced standards.

- 5.76 Foam discharge shall be managed to prevent environmental contamination. The testing of foam suppression systems must be conducted with no release to the environment, including storm water or sewerage systems.

Note: Defence prefers the use of the water equivalency method for commissioning and maintenance testing where this meets the requirements of the referenced standards.

Emergency planning and training

- 5.77 Emergency planning in Defence facilities must be compliant with the referenced standards and the Defence Base Emergency Management and Incident Response.

FIRE SAFETY SURVEYS (FSS)

Background

- 5.78 FSS are a vital component of fire safety management within Defence estate buildings. Different states in Australia have different requirements for FSS. The Defence FSS provides a standardised approach that addresses all the relevant state and statutory requirements, along with those specific to Defence.
- 5.79 The Defence FSS program is designed to:
- a. evaluate the performance of fire safety systems
 - b. ensure compliance with fire engineering standards
 - c. identify areas for improvement.
- 5.80 FSS assist in mitigating fire risks and ensuring the protection of both building occupants and Defence assets. They help to identify maintenance issues and support programming of fire safety works across the Defence estate.

General requirements

- 5.81 The elements of the NCC that must be assessed in the FSS include the minimum standards necessary to ensure the safety of persons in a fire event by confirming their adequacy of:
- a. fire compartmentation and separation
 - b. egress provisions
 - c. services and equipment
 - d. fire safety measures performance.
- 5.82 The FSS shall be conducted for all buildings owned, leased, or occupied by Defence.
- Note:** The FSS product director may choose to exclude buildings where it is considered that the conduct of a fire safety survey is not warranted. This may include but is not limited to CF3 to CF5 buildings that are class 5-9.

Frequency of the surveys

- 5.83 The frequency of FSSs shall be determined and recorded against the structure in the Garrison Estate Management System. Defence buildings subject to a FSS requirement are to be inspected at a minimum of once every 3 years.
- 5.84 FSSs shall be conducted:
- a. annually for CF1 buildings

- b. every 3 years for CF2 – CF5 buildings.

Reporting and inspection requirements

- 5.85 Each FSS shall result in a detailed report outlining findings, identified issues, and recommended corrective actions. The reports shall follow a standard template provided in the guidelines for fire safety surveys.
- 5.86 A fire safety survey is a walk-through visual inspection to assess the effectiveness the buildings fire safety measures. Access to roof and floor spaces may be required to determine compliance of fire safety provisions.
- 5.87 Functional testing of fire systems and equipment is the responsibility of the maintenance contractor and is not part of this survey, however it may be requested if considered necessary.

Authorised inspecting officer

- 5.88 Fire safety surveys shall be conducted by qualified and accredited inspecting officers who possess the necessary certifications to carry out inspections. Inspecting officers must hold one of the following qualifications:
 - a. accreditation/ licensed as a building surveyor in the relevant state or territory
 - b. accreditation under a recognised Australian building surveying accreditation scheme, such as that offered by the Australian Institute of Building Surveyors.

Note: Inspecting officers must only perform assessment works within the level of their accreditation/ license restrictions and comply with Defence security protocols and obligations.

Governance and responsibilities of FSS Program

- 5.89 Service Delivery Division is responsible for ensuring that all aspects of the fire safety survey program are managed and executed effectively across the Defence estate.

Conflict of Interest

- 5.90 The potential exists for a conflict of interest if there is a direct link between maintenance agents and parties conducting FSS. Conflicts of interest between parties undertaking FSS and those operating and maintaining the estate must be identified and managed effectively.

Fire safety survey report (FSSR) structure guidance

- 5.91 For FSSR structure, refer to Guideline 5002.

Fire safety survey recommendations

- 5.92 FSSR recommendations shall outline actions required to address compliance and or conformance, maintenance, and management issues. Where further assessment is recommended, the inspecting officer shall specify the required discipline along with the defined scope and urgency of the assessment, including applicable timeframes.
- 5.93 The inspecting officer may consider recommending a NCC/manual audit owing to identification of significant compliance, conformance or certification issues.

Work, health and safety (WHS) risks

- 5.94 During FSS, any identified WHS risks must be immediately addressed. Where the safety hazards are identified, a WHS hazard and risk assessment must be conducted and appropriate actions taken by the inspecting officer and program delivery authority.

BUILDINGS FOR DISPOSAL

Compliance requirements

- 5.95 This section details minimum levels of fire protection and fire safety measures which must be maintained until disposal is completed.
- 5.96 Any building that is identified for disposal and is still in use must comply with all the requirements of the fire engineering section of this manual and the relevant codes in force at the time of construction.
- 5.97 Any building that has a change in use must comply with the requirements for the new use as set out in this manual under building works section.
- 5.98 Buildings identified for disposal that are vacated must as a minimum comply with the following:
- a. where installed, retain and maintain fire hydrants, fire detection, sprinklers and emergency warning systems
 - b. unauthorised access must be prevented
 - c. exit doors must be able to open and exit paths must be kept clear.

Note: Other fire safety measures or those that were installed solely to meet the fire engineering requirements of this manual may be decommissioned where safe to do so.

Duty of care

- 5.99 Buildings deemed for disposal may not have enough personnel to take immediate action in a fire. A building scheduled for disposal is still required to comply with relevant legislative requirements.
- 5.100 Duty of care obligations require Defence to take a responsible approach when preparing buildings for disposal, so as not to unduly endanger the occupants or property of neighbouring establishments, or to place firefighters in unnecessary danger.

FIREFIGHTING WATER

General requirements

- 5.101 A firefighting water supply, either separate or integrated with potable water, must be available on the Defence estate where an automatic fire sprinkler, fire hydrant, fire hose reel, or other water-based fire suppression system is installed. Firefighting water supplies are to comply with the requirements of the NCC and related Australian Standards in force at the time of building approval and operational requirements of the responding fire service. Detailed requirements are included for fire systems in the Water Source and Reticulation [Guideline 5013](#).

Water sources and the adequacy

- 5.102 The water quality and reliability shall be suitable for the operation of the fire protection systems, with consideration for compatibility with fire protection system materials, including the chemical and biological composition and pH levels.
- 5.103 Where a water source other than an authority reticulated supply is used, such as an inexhaustible supply, evidence of the reliability and suitability of the source must be provided in line with the relevant Australian Standards.

5.104 The designer must coordinate with maintenance contractors to review water infrastructure maintenance records and assess whether the existing system can support the proposed new connections.

5.105 Where deficiencies are identified in the existing water infrastructure, consideration shall be given to upgrading the infrastructure to ensure compliance with operational requirements.

Where practical and viable for Defence, the firefighting water infrastructure and reticulation shall be separate from potable water.

Requirements of CF1 and CF2 Facilities

5.106 CF1 and CF2 facilities shall have dual water supply sources, with each source capable of providing the required system duty (pressure and flow) for the required duration in accordance with relevant standards.

5.107 At least one of the water supply sources shall be from a full capacity static water storage tank or inexhaustible water supply. The tank capacity for one of the sources is not permitted to be reduced via automatic infill.

Water Authorities

5.108 Consultation with the local water authority must be undertaken for all works where the firefighting systems interface to an authority main water supply. The designer must keep the water operating authority informed of the consultation and outcomes.

5.109 As a minimum, the consultation is to consider the available pressure and flow from the town's main water supply and its associated availability and reliability.

Sustainability

5.110 The fire water supply design must incorporate measures to reduce the use (wastage) of water during routine testing and maintenance and include consideration for options which minimise whole of life emissions from the system.

Defence Specific Requirements

5.111 Guidelines articulating Defence specific requirements relevant to this chapter are:

- a. 5001 – Fire Safety Management Strategy (FSMS)
- b. 5002 – Fire Safety Survey (FSS)
- c. 5003 – Computing and electronic equipment, CF1 and CF2
- d. 5004 – Storage and workshop buildings, CF1 and CF2
- e. 5005 – Aircraft hangars
- f. 5006 – Building construction in bushfire prone areas
- g. 5007 – Explosive ordnance buildings
- h. 5008 – Defence living-in accommodation
- i. 5010 – Enclosed carparks of electrical-vehicles and lithium ion storage battery facilities
- j. 5011 – Decommissioning a non-required system
- k. 5012 – Special structures
- l. 5013 – Water Source and Reticulation

CHAPTER 6

ELECTRICAL ENGINEERING

GENERAL PRINCIPLE

- 6.1 The design of Defence electrical systems shall ensure that the system can be constructed, operated, maintained and decommissioned in a way that minimises risk or hazard to persons, property, external organisations, the community or the environment.

Rationale

- 6.2 The rationale of this chapter is to reinforce that all Defence electrical systems must always:
- a. be in full compliance with all applicable environmental, legislation, regulatory and applicable standards
 - b. adhere to the requirements and obligations set out in the applicable NSP's service installation rules and any connection agreement in effect.

Expected outcome

- 6.3 Electrical systems and its infrastructure shall comply with all relevant state legislation, the NCC, and applicable Defence policies and standards governing the design, construction, modification, and maintenance of building services and infrastructure.

General requirements

- 6.4 The NDS 2026 directed that Defence must accelerate its transition to clean energy to improve base energy resilience. Defence is committed to accelerating its transition to clean energy across Defence estate.
- 6.5 This acceleration is essential to strengthening energy resilience in support of Defence capability and continuity of operations, improving energy efficiency and reducing emissions. Achieving these outcomes requires the systematic and risk-informed integration of renewable energy generation systems, energy efficiency measures, and enabling technologies, in accordance with Defence policy and national energy frameworks.
- 6.6 All projects and persons must use their best endeavours to ensure Defence electrical systems are as far as possible directed towards this purpose, increasing on-site clean energy generation, storage, and electrification where possible.
- 6.7 The planning of Defence electrical infrastructure systems should be undertaken via one or a combination of:
- a. Electrical Master Plan – EMP
 - b. Electrical Development Plan – EDP
 - c. Electrical Network Systems Report – ENSR.

Note: The structuring, format and revision issue for EMP, EDP & ENSR are addressed in more detail in Guideline 6001 – Planning Reports. Further requirements for planning clean energy systems are set out in 6.79-6.87.

- 6.8 Access to, operation of, or work in proximity to the Defence electrical distribution system is strictly prohibited without prior authorisation from the Electrical Operating Authority (EOA). This includes areas:
- a. in proximity to underground electrical infrastructure
 - b. housing high voltage (HV) equipment
 - c. considered as work zone B or C within the work safe Australia Guidelines (example figure 6.1 below).

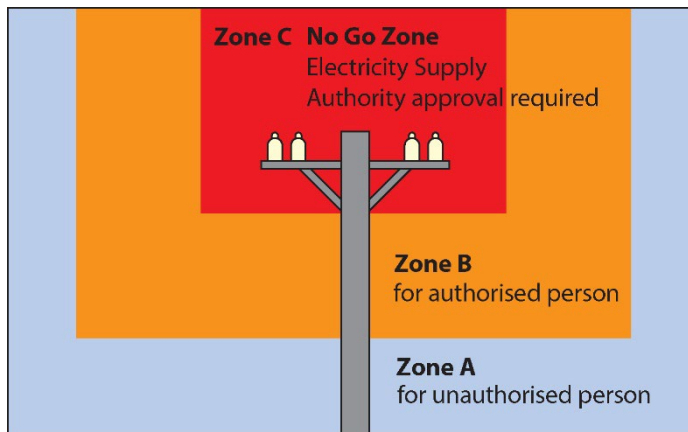


Figure 0.1: Work zones in the vicinity of overhead electric line poles

ELECTRICAL OPERATING AUTHORITY

- 6.9 Defence, as the owner and operator of electrical distribution systems at its facilities, retains overall responsibility for their management. Technical oversight and operational safety of each base electrical distribution system are delegated to the EOA. These services are typically delivered by the Property and Asset Services (PAS) contractor under the Base Services Contract. Under certain operating conditions, Defence personnel may assume responsibility for some or all of these functions.

EOA responsibilities

- 6.10 The EOA oversees the daily operation, maintenance, and management of the Defence electrical distribution systems. Its responsibilities include:
- a. network operation, which involves:
 - i. ensuring a continuous supply of electricity from the distribution system to users
 - ii. managing the configuration and operation of the electrical distribution systems under both normal and abnormal conditions
 - iii. implementing documented procedures for operating the electrical distribution system.
 - b. regulating access to the electrical distribution systems, which includes:
 - i. authorising personnel requiring access to, operate, or work on the electrical distribution system

- ii. ensuring the safety of all personnel accessing or working near the distribution system, including issuing access permits for activities within safe working distances
 - iii. maintaining compliance with existing Power Purchase Agreements (PPA) and other contractual arrangements with Defence.
- c. managing the status and condition of the Defence electrical distribution system involves:
 - i. keeping documentation up to date regarding the configuration of the Defence electrical distribution system
 - ii. monitoring the condition, status, and loads within the Defence electrical distribution system
 - iii. accepting and integrating new infrastructure into the electrical distribution system
 - iv. ensuring that any new work, upgrades, or enhancements to the electrical distribution systems comply with this policy
 - v. maintaining plant and equipment in accordance with the base services contract requirements, as well as all relevant legislation and regulations.

Acceptance of network connections

- 6.11 The Defence EOA or appointed representative reserves the right to deny any connection or energisation of works to the electrical distribution system if the following conditions are met:
- a. the works compromise safety or fail to fully comply with relevant state and territory legislation and regulations. This must be substantiated by a formal safety and risk assessment that evaluates the works against:
 - i. applicable codes, standards, and guidelines
 - ii. legislative requirements.
 - b. the works compromise operational integrity or do not align with this manual. This must be supported by a formal review and risk assessment that evaluates the works against:
 - i. this manual
 - ii. any brief or directive from Defence regarding the works, acknowledging that Defence may authorise works that do not conform to the above standards
 - iii. the necessary updated documentation and records, as required by this manual, have not been provided, or the updated network schema is missing.
 - c. If the EOA or authorised representative denies connection or energisation, it must submit a formal report to the project sponsor.
- 6.12 This report will help the project sponsor decide on a further course of action. Additionally, the report will include a formal review and risk assessment that evaluates the work in relation to the relevant items a, b, and c mentioned earlier.

Support for projects

- 6.13 The EOA shall provide reasonable support to projects undertaking works upon the electrical distribution systems. This shall include:
- a. any required network switching and isolation
 - i. provision of system information including loads and spare capacity.
- 6.14 Where the requirements for this support may be beyond the resources readily available from Authorised Personnel for the base, the EOA may come to an agreed arrangement with a project. This could include and may not be limited to:
- a. payment for required overtime
 - b. payment for temporary employment of additional authorised personnel
 - c. authorisation of the project's personnel.

System outages and resilience

- 6.15 The EOA shall ensure that where the HV system is isolated in whole or part, that the duration of any power interruption or reduction in redundancy is as short as practical. This includes isolations that affect the ability to move the open points on rings.
- 6.16 Before any power interruption or reduction in redundancies occurs, the relevant Head of resident unit (HRU) and base manager must be consulted.

New and existing network connections

- 6.17 The design and construction of new or modified facilities, equipment, and connections is a complex undertaking that, if improperly executed, may introduce significant risks to health, safety, security, operational continuity, environment compliance, and system reliability. To mitigate these risks and prevent unforeseen delays, coordination with the EOA is mandatory throughout all stages of planning and implementation.

Documentation and records

- 6.18 The EOA shall maintain all required documentation and records pertaining to the Defence electrical distribution systems. These records must be stored in an agreed, easily accessible format within the Defence database, as determined in consultation with the Directorate of Estate and Land Management (DELM) for each base. Where a project involves modifications to Defence electrical distribution systems, the project is responsible for preparing and submitting updated documentation to the EOA as part of the handover–takeover process.
- 6.19 The EOA shall be responsible for the management and maintenance of all components of the Network Configuration manual. In the event that any required documentation listed within the manual is missing or incomplete, this deficiency must be reported promptly to the relevant Defence contract manager:
- a. operating protocol between Network Service Provider and Defence
 - b. network service agreements between Network Service Provider and Defence
 - c. a network geographic plan showing the general location of the electrical distribution network across the base
 - d. HV and Low Voltage (LV) Schematics and single line diagrams for each substation, switching station and LV Distribution Pillars

- e. for LV services: Distribution board, Distribution pillar (overhead or LV network) building
 - f. for LV services: Street lighting on the base along with associated distribution boards/pillar locations
 - g. location of EV charging
 - h. location and specifications of generation systems (Local Emergency Generators (LEGs), Central Emergency Power Station (CEPS), Central Power Station (CPS), renewable energy systems)
 - i. protection co-ordination report for the entire system, and all protection relay settings files and protection settings for the base
 - j. earthing reports and the results of earth grid testing for each substation and switching station
 - k. arc flash studies
 - l. load flow studies for the entire network.
 - m. base network operating procedures (access, electrical operating and permit procedures for the base)
 - n. underground cable locations across the base, including pit and pipe systems.
 - o. overhead line drawings – (inclusive of pole locations, conductor type, span and fittings)
 - p. Network Safety Management Plan and contact details for the appointed Network Controller, and authorised operating personnel
 - q. bushfire risk management plan and associated risk assessments
 - r. operating protocols for abnormal operating conditions e.g. loss of primary power supply
 - s. any other management plans or information pertaining to the operation of the electrical distribution network
 - t. hazardous area dossiers
 - u. logbooks
 - v. electrical distribution system elements inspection and maintenance plan(s)
 - w. operation and maintenance manuals.
- 6.20 All documentation and records referenced above shall be maintained in an up-to-date condition and accurately reflect the physical configuration and operational performance of the electrical distribution systems at the base. These records must be stored within the Defence database, as agreed with DELM, in both:
- a. editable raw data format for ongoing updates and revisions
 - b. PDF format for controlled access by Defence personnel, contractors, consultants, and other authorised parties
- 6.21 The EOA shall ensure these documents are made available as required or upon formal request from any project intending to undertake work on the electrical distribution systems.

Authorisation to access or work upon or in close proximity to Defence networks

- 6.22 The EOA shall be the sole entity to accredit and grant permission for individuals to access, operate, or perform work on, or in close proximity to, Defence electrical distribution systems.
- 6.23 All authorisations issued by the EOA shall fully comply with all applicable legislative and regulatory requirements for the relevant jurisdiction.
- 6.24 No individual shall access, operate or work upon, or in close proximity to, the Defence electrical distribution systems without explicit authorisation of the EOA.
- a. Close proximity relates to areas housing HV equipment or in proximity to earthing systems of HV systems where those earthing systems are likely to become exposed due to the works.
- 6.25 All personnel and equipment shall maintain a minimum clearance of 3 meters from any uninsulated and exposed components of the Defence electrical distribution system at all times, unless explicit authorisation has been granted by the EOA or authorised delegate.
- a. Examples of uninsulated and exposed parts of the electrical distribution networks are:
 - i. bare conductors and cabling on overhead power lines
 - ii. exposed power cable terminations or joints within substations, switching stations, pillars or pits.
- 6.26 The EOA shall maintain documented processes and procedures that define the levels and categories of accreditation and authorisations for all activities involving Defence electrical distribution systems. The process shall include, at a minimum:
- a. the levels of experience and competencies required for each category of authorisation
 - b. the methods for assessing experience and competencies
 - c. training requirements, including initial, ongoing and refresher training
 - d. procedures for renewing authorisations
 - e. systems for recording, maintaining and updating authorisation records.
- 6.27 The EOA shall ensure that:
- a. authorised personnel possess requisite competency and experience for tasks assigned
 - b. the scope, level, and limitations of each authorisation are clearly defined, communicated, and acknowledged by the authorised individual
 - c. authorisations are issued for specific, finite time periods aligned with the nature and duration of the work
 - d. authorisations are revoked immediately when no longer required, with formal acknowledgement by the authorised individual
 - e. the issuance, acceptance, and cancellation of all authorisations are formally documented and recorded.
- 6.28 Prior to commencing any works where personnel or activities may come into close proximity to overhead power lines or may expose or operate near underground power cables, the EOA

or designated site delegate must be consulted. All directions and guidance issued by the EOA or their delegate shall be strictly adhered to.

POWER GENERATION SYSTEMS

- 6.29 The connection of any power generation system onto a Defence electrical distribution system shall be in full accordance with this manual. Power generation systems may be located on a Defence establishment for some of the following reasons:
- a. as a source of normal supply, generally when the mains is unavailable at the site, or has insufficient capacity or quality
 - a. to provide standby power in the event of loss of normal power supply. This is referred to as an embedded generation/ standby power system or emergency generator
 - b. to support the normal supply during periods of peak electrical demand (this is referred to as peak lopping)
 - c. to offset energy consumption from the normal power supply (this often involves the use of renewable energy sources)
 - d. to address power quality issues with the normal power supply, for instance voltage regulation and stability issues
 - e. specialist systems require power at different frequencies from the normal power supply and such circumstances require power conversion systems.
- 6.30 Power generation systems must:
- a. use renewable energy and deliver required energy to a base unless there is a compelling reason not to do so. This may be through, for example, onsite energy generation and storage systems, such as through a solar and battery energy storage system
 - b. avoid additional emissions and not generate greenhouse gas emissions unless there is a compelling reason to do so.
 - c. consider whether the required power for a base may be delivered through installing or using electricity demand reduction measures to reduce existing power consumption.

Types of generation systems on Defence establishments

- 6.31 Power generation technologies currently installed on Defence establishments include:
- a. liquid or gas fuelled generating units utilising reciprocating engines or gas turbines
 - b. Fuel Cell Systems
 - c. renewables, such as solar or wind
 - d. Battery Energy Storage Systems (BESS)
 - e. combined power generation with co-generation or waste heat recovery for additional services, such as air conditioning.
- 6.32 Generation systems on Defence bases are classified as either systems connected at HV to the Defence network, or systems connected at LV to the Defence network.

Note: Any proposed generation system on a Defence establishment shall meet the NSP requirements. Appropriate studies shall be conducted before connecting to the Defence electrical network. All systems shall be treated as large generator.

- 6.33 The type of generation system provided shall comply with the requirements specified in the requirement of standby power.

System connected at HV to the Defence Network

- 6.34 Typically connected at HV to the Defence electrical distribution system, such as:
- a. CPS: A power station with a principal purpose to provide normal power to an establishment, or large portions of an establishment. This can be as the sole source of power or in conjunction with other sources, such as the grid or another CPS
 - b. CEPS: A power station with a principal purpose to provide standby power to an establishment, or large portions of an establishment, in the event of failure of the normal supply. Like a CPS, the power generation technology adopted at CEPS can be quite varied and even incorporate hybrid solutions such as Diesel Rotary Uninterruptable Power System (DRUPS) and renewable energy systems combined with batteries and backup generators
 - c. Central Power Conversion Station (CPCS): A power station with a principal purpose to change the frequency of the incoming electrical supply to provide power for specialist applications, such as shore power, to an establishment or large portions of an establishment.

System connected at HV to the Defence Network – Defence-owned

- 6.35 These requirements apply to large, centralised power generation that is owned, operated and/or maintained by Defence for which Defence is responsible or liable.
- 6.36 Generation assets shall be provided in a manner that:
- a. allows the generation plant to operate at full capacity, safely and reliably under any reasonable weather condition at the location, subject to any higher requirement specified by the capability manager
 - b. is survivable under extreme weather conditions at the location, so that the plant can be quickly brought into service after such an event
 - c. considers passive defence and is consistent with the required design life (this normally requires that the plant is housed within a sturdy building)
 - d. prevents excessive degradation of the assets due to environmental conditions at the location
 - e. allows maintenance activities to be undertaken under any reasonable weather condition at the location
 - f. considers the maintenance support, including timely supply of spare parts, that is available at the location
 - g. provides adequate maintenance access so that the plant can be safely accessed, even when running

- h. meets all environmental requirements, including containment of spills, such as fuel, lubricants or electrolytes; noise limits and gaseous or exhaust emissions
 - i. considers future expansion of the installation
 - j. has suitable segregation between major components to minimise the potential for a single event to disable the entire station
 - k. be provided with the amenities to support the operation and maintenance of the facility
 - l. enables operators to be able to visually monitor the generators remotely i.e. without physically being in the generator hall
 - m. provides a consistent operational, control and monitoring system across the estate and the base be able to operate uninterrupted for an agreed minimum period without the need for refuelling, maintenance and servicing.
- 6.37 Emergency and/or stand by systems shall be capable of restoring supply automatically and autonomously on the loss of any incoming supply.
- 6.38 Defence owned large centralised power systems shall be fully integrated into the base's HV network and its monitoring, control and operational systems.

System connected at HV to the Defence Network – third party-owned

- 6.39 These requirements apply to large, centralised power generation that is owned, operated and or maintained by third parties, but which supply power into a Defence electrical distribution system.
- 6.40 The design, construction, operation, maintenance and management of such systems shall always be in full compliance with:
- a. all applicable legislative, regulatory and NSP requirements
 - b. contractual requirements between Defence and the third party.
- 6.41 The connection and operation of any such system shall:
- a. not jeopardise or present a threat to the resilience, reliability, operation and/ or security to the electrical distribution system or the NSP supply
 - b. meet legislative, regulatory and NSP requirements without placing future restrictions on development of Defence systems (e.g. Defence power generation systems) without the prior specific approval of Defence
 - c. provide the supply availability defined in the contractual requirements between Defence and the third party.
- 6.42 Where generators are provided as the primary power supply for a Defence establishment the availability of the system should generally be no less than 99.99%.
- 6.43 Defence shall always have the authority to disconnect, physically or electrically, a system from the Defence network where it considers it:
- a. is adversely affecting power system reliability, stability or power quality to Defence or at the point of connection to the NSP, such that these fall outside acceptable limits
 - b. poses a threat or safety risk

- c. is in breach or non-compliant with any applicable legislative, regulatory or NSP requirement
 - d. is in breach of the contractual requirements between Defence and the third party or any other Defence contractual obligations.
- 6.44 Point of Connection - the connection of any such systems shall be in full accordance with an agreed Electrical Master Plan:
 - a. connections should be made at the primary nodes of the HV network. For example ISS, PSS, CEPS, etc.
 - b. the point of connection shall be via fully automated circuit breaker which shall be able to be:
 - i. monitored, opened and closed by Defence
 - ii. monitored and opened by the third-party operator. The third-party operator shall not be able to close the breaker.
- 6.45 Defence shall always have full visibility and be able to monitor and capture and retain at the connection point the following:
 - a. frequency
 - b. volts – each phase
 - c. current – each phase
 - d. power factor
 - e. real power kW
 - f. apparent power kVA
 - g. reactive power kVars
 - h. kWh at 5-minute incremental periods and cumulative for the 24 period.
- 6.46 Under no circumstances shall any third-party owner, operator, maintainer or their representative be granted access to, visibility of, or control of the Defence Electrical network or its systems unless otherwise explicitly permitted by this policy.
- 6.47 Where a system or portions of the system is to revert to the ownership of Defence at any point in time, the relevant portions of that system shall be in full conformance with the requirements of this policy.

System connected at LV to the Defence Network – small embedded power generation systems

- 6.48 Small embedded power generation systems shall only be connected to the Defence electrical distribution system following:
 - a. consultation with responsible EOA regarding any power system limitations
 - b. performance of a power system study that determines the normal and worst-case effects on power quality from the proposed embedded generation and any existing embedded generation at:

- i. the installation
- ii. the source of supply, e.g. the distribution transformer
- iii. any other loads connected to the same source of supply.

- 6.49 The results of this study shall inform the capacity of the embedded generation that is installed.
- 6.50 Small embedded generation systems shall be provided with automatic protection to immediately disconnect from the Defence electrical distribution system in the event of a loss of normal supply and shall remain disconnected until supply to the Defence electrical distribution system is restored and stable.
- 6.51 Where small embedded generators are used to supplement local standby power systems, any switchboard supplied by the embedded generator, either directly or otherwise, shall be clearly labelled.

System connected at LV to the Defence Network – small distributed standby power generation systems

- 6.52 These requirements apply to small distributed standby power generation systems, typically connected at LV to the Defence electrical distribution system.
- 6.53 Typically connected at LV to the Defence electrical distribution system, such as:
- a. local emergency generators: a generator or number of generators with the principal purpose of providing power to facilities, or portions of a facility, in the event of a failure of the normal supply
 - b. uninterruptable power supplies: an energy storage and supply system to maintain supply electrical energy to a facility in the event of a short-term power outage and is typically connected in series with primary power supplies
 - c. hybrid solutions such as renewable energy microgrids consisting of solar PV with a BESS or solar PV, BESS and diesel generator combining the functions of LEG and UPS in a single unit or a diesel rotary uninterruptable power system that combines the functions of a LEG and UPS in a single unit
 - d. mobile generator link box: facilities for connection of transportable generating sets.

Local emergency generators (LEG)

- 6.54 The LEG shall be capable of automatic and autonomous operation to:
- a. provide standby power on the loss of the normal supply as soon as reasonably practical following the loss of that supply. The transfer to LEG supply shall be break before make
 - b. the load applied to the LEG while it is operating shall remain above the manufacturer's minimum load. Where the actual load falls below this minimum load additional resistive load shall automatically be added to keep the load above the minimum
 - c. on restoration of normal supply, following a suitable time delay to ensure stability of the normal supply, the transfer to incoming supply shall be on a break before make basis.
- 6.55 Manual controls shall be provided to allow the manual operation of the above functions.

- 6.56 Remote operation and monitoring shall be provided where the LEG needs to integrate with other Defence systems, such as CEPS and Air Traffic Control. Remote control overrides the automatic starting and stopping of the LEG. This typically includes:
- a. remote start (and connect) and stop (after disconnecting)
 - b. LEG run-on, where the LEG will operate as if the normal supply is not present
 - c. status, including run, stop, connected, fault etc.
- 6.57 LEG fuel systems have adequate capacity to support the LEG operating at the prospective load until it can be appropriately refuelled. This can be augmented via additional storage or integration with renewables and BESS to function as a building-level microgrid. Further guidance on an appropriate period is provided in Guideline 6003.
- 6.58 Generation assets shall be housed in a manner that:
- a. allows the generation plant to operate at full capacity, safely and reliably under any reasonable weather condition at the location, subject to any higher requirement specified by the capability manager
 - b. can be monitored and operated by semi-skilled personnel
 - c. is survivable under extreme weather conditions at the location, so that the plant can be quickly brought into service after such an event
 - d. prevents excessive degradation of assets due to environment conditions at the location
 - e. allows maintenance activities to be undertaken under any reasonable weather condition at the location
 - f. considers the maintenance support, including timely supply of spare parts, that is available at the location
 - g. provides adequate maintenance access so that the plant can be safely accessed, even when running
 - h. meets all environmental requirements, including containment of spills and gaseous emissions and noise limits
 - i. considers future expansion of the installation and integration with renewables such as solar PV and BESS
 - j. maintains a consistent operational, control and monitoring system across the estate and base
 - k. can operate uninterrupted for an agreed minimum period without the need for refuelling, maintenance and servicing.

Uninterruptable power supply (UPS)

- 6.59 UPSs shall be arranged to enable maintenance without interruption to equipment/facilities the UPS supports. This may be achieved through modular UPS systems or external bypass systems.
- 6.60 UPSs shall be compatible and suitable for the load it is supporting, including where a LEG may also be providing further emergency/standby power support.

6.61 UPSs shall be housed in a manner that:

- a. allows the UPS to operate at full capacity, safely and reliably under any reasonable weather condition at the location, subject to any higher requirement specified by the capability manager and including extended loss of normal power supply
- b. can be monitored and operated by semi-skilled personnel
- c. is survivable under extreme weather conditions at the location, so that the plant can be quickly brought into service after such an event
- d. prevents excessive degradation of assets due to environment conditions at the location
- e. allows maintenance activities to be undertaken under any reasonable weather condition at the location
- f. considers the maintenance support, including timely supply of spare parts, that is available at the location
- g. provides adequate maintenance access so that the energy storage system can be safely accessed, even when running
- h. meets environmental requirements, including containment of spills and gaseous emissions
- i. considers future expansion of the system.

Renewable energy hybrid solutions

6.62 Renewable energy hybrid solutions include microgrids consisting of solar PV with a BESS or solar PV, BESS and diesel generator combining the functions of LEG and UPS in a single unit.

Mobile Generator Link Box (MGLB)

6.63 The objective of a MGLB is to:

- a. facilitate the safe and convenient connection of mobile generators
- b. provide a consistent mobile generator connection interface.

6.64 The provision, in design and construction, of a MGLB across the estate shall be of a consistent and uniform configuration and standard.

Diesel Rotary UPS (DRUPS)

6.65 DRUPS shall comply with the relevant provisions for both UPS and LEGs.

6.66 LEGs and DRUPS shall be suitably sized to support the load required, and when newly installed have spare capacity to cater for any potential future load growth.

Requirement of standby power

6.67 Standby power shall be provided where electrical power is required to maintain critical and essential capabilities upon failure of the normal supply, and where there are no practical alternative ways of maintaining that capability without serious degradation of the capability, e.g. manual operations rather than automatic.

- 6.68 Critical and essential capabilities are generally those classified as contribution factor CF1 and CF2. In general, standby power systems should be located close to the facility in which the capability is located.
- 6.69 The selection of standby power system shall be based on the characteristics of the connected load, particularly the maximum allowable power supply interruption before operational capability is significantly compromised. The standby power system will typically include one or more of the following components:
- a. no permissible power interruption: typically, computer systems where data loss will compromise capability or for safe shutdown requirement - UPS
 - b. zero to 60 seconds permissible power interruption requirement - LEG
 - c. more than one-day permissible power interruption in locations where standby generators can be hired at short notice - MGLB
 - d. renewable and hybrid solutions, such as DRUPS, and microgrids consisting of solar PV with a BESS or solar PV, BESS and diesel generator combining the functions of LEG and UPS in a single unit. These systems can also be dispersed while providing the same benefits.
- 6.70 The level of redundancy provided in the standby power system shall consider:
- a. the importance of the capability
 - b. the reliability of the normal electrical supply.
- 6.71 Where required, redundancy within the standby power system can be provided by either:
- a. providing redundant generating units and ancillary systems
 - b. provision of a CEPS.
- 6.72 Establishment of a CEPS shall require that the justification is set out within an agreed and promulgated Electrical Master Plan. Typical justifications include:
- a. the establishment has a post disaster functionality and requires that power be broadly available for the efficient delivery of this functionality
 - b. the normal electrical supply is of poor reliability or quality, and this affects the overall operation of the establishment
 - c. additional redundancy is required for individual standby power systems.

CLEAN ENERGY ACCELERATION

- 6.73 This section sets out the principles, strategies, and generation options that guide clean energy implementation across the Defence Estate. It enables consistent, risk-informed integration of clean energy systems into Defence infrastructure; improves decision-making through clear, actionable guidance; and enhances the management of technical, operational, and security risks associated with clean energy deployment. Collectively, these measures support operational resilience, self-sufficiency, and long-term sustainability objectives across the Defence Estate.
- 6.74 For most bases in Australia, on-base renewable energy represents a cost-effective method for Defence to procure energy. Renewable energy systems typically involve high up-front capital costs, followed by low or negligible ongoing operating costs. Their main primary financial benefit is measured through the 'payback period', which represents the time

required for that system to generate enough 'cost-free' energy to offset its installation and maintenance expenses.

- 6.75 When successfully deployed, on-base renewable energy will supply cost-effective energy security improvements which contribute directly to on-base capability as well as legislated targets. This outcome will be especially valuable during impacts to liquid fuel energy supply chains which could be expected during competition or conflict scenarios.

Foundational requirement

- 6.76 Projects that seek to install power generation systems in a Defence-owned or leased building or facility must use their best endeavours to avoid technologies that generate greenhouse gas emissions during normal operation while maintaining energy resilience and reliability. Preference should be given to low or zero-emission generation sources such as solar or battery storage.
- 6.77 Projects must make every reasonable effort to avoid including, where possible, embedded power generation systems that generate greenhouse gases, such as diesel or petrol generators, while maintaining energy resilience and reliability. This specifically means projects must avoid installing diesel generators wherever possible.
- 6.78 All projects that include a clean energy system shall:
- a. design and construct renewable energy systems with appropriate equipment (such as metering and telemetry) to ensure eligibility for accreditation to create generation certificates under the Renewable Energy (Electricity) Act 2000 and associated regulations
 - b. while noting that the Resource Data Management System (RDMS) has now been decommissioned, install sub-metering on all solar distributed energy assets in accordance with National Electricity Market (NEM) and National Electricity Rules (NER) requirements.

Renewable energy integration planning

- 6.79 Bases must consider the cumulative effect of on-base renewable generation. The key risks from on-base renewable generation are cumulative harmonic distortion from inverters, reverse power flow through sub-stations or connections to an external civilian network, dynamic instability or cascading failure across the site, external network integration requirements and management of energy sources during scheduled and un-scheduled outages and maintenance.
- 6.80 The integration plan shall include as a minimum:
- a. harmonic distortion assessment (desktop assessment may be sufficient for smaller systems)
 - b. load flow assessment
 - c. protection and fault current assessment
 - d. control scheme proposal (distributed control compliant with AS 4777.2 may be sufficient)
 - e. proposal demonstrating compliance with DNSP.
- 6.81 A renewable energy integration plan shall be created and maintained for all:

- a. bases with more than 3 separated renewable energy systems above 20 kVA connected to the same internal electrical network
- b. bases where installed capacity exceeds 20% of the highest recorded peak demand
- c. bases where local DNSP requires monitoring/control of on-base renewable energy systems.

Integration requirements

- 6.82 Renewable energy sources bring unique challenges to electrical power systems when integrated into the grid.
- 6.83 Renewable energy systems shall be integrated with existing energy systems through appropriate assessment and planning of the base electrical network.
- 6.84 As a minimum, renewable energy systems shall:
- a. be incorporated into any existing PCMS, including SCADA
 - b. provide wired communication facilities capable of implementing at least one industry standard communication protocol
 - c. include the capability to cease generation, either through inverter remote control, microgrid controllers, or dedicated relays
 - d. comply with DNSP requirements.
- 6.85 When assessing Clean Energy projects key technical, operational and assurance considerations must be considered, including:
- a. siting requirements
 - b. glint & glare impacts
 - c. electromagnetic interference
 - d. power quality
 - e. cybersecurity
 - f. operation & maintenance.
- 6.86 Queries related to clean energy projects across Defence should be directed to: cleanenergy.transition@defence.gov.au.
- 6.87 Guideline 6006 provides further information on the Clean energy acceleration requirements.

ENERGY STORAGE SYSTEMS

- 6.88 All energy storage systems shall meet applicable regulatory and legislative standards.

Battery Energy Storage Systems (BESS)

- 6.89 A BESS is designed to store energy generated by power sources. When designing energy systems, consideration shall be given to the installation of BESS to enhance reliability, support load management, and improve overall energy efficiency. In addition, the following objectives should be addressed:
- a. maximise consumption of solar photovoltaic (PV) generation behind the meter

- b. minimise PV generation curtailment that may be imposed by a Distributed Network Service Provider (DNSP) during grid-connected operation
- c. improve power quality and maintain energy security when operating in 'islanded' mode
- d. increase the estate's capability to manage its energy independently
- e. stabilise the energy grid during periods of peak demand.

6.90 Provide functionality as a UPS and/or complement existing backup power sources.

SYSTEM STUDIES

6.91 The power system studies are conducted to ensure the electrical network operates safely and reliably under both normal operating conditions and during fault scenarios.

Power system stability and quality

- 6.92 A power system must remain intact and capable of withstanding a wide variety of disturbances.
- 6.93 Power system stability and power quality within the Defence electrical distribution system shall be maintained within acceptable limits at all times, to the extent influenced by the power generation system. The following requirements apply:
- a. Voltage and Frequency: Voltage and frequency shall remain within the limits specified by applicable Australian Standards or the Network Service Provider (NSP), under all operating conditions from no-load to full load.
 - b. Flicker and Harmonics: Flicker caused by large intermittent loads or generating systems shall comply with Australian Standards and NSP requirements and shall not adversely impact other users or equipment. The harmonic content of the electrical supply shall also comply with Australian Standards and NSP requirements, considering both existing harmonics and any new sources introduced.
 - c. Impact on Users and Stakeholders: Harmonics and other power quality issues shall not cause adverse impacts on other users, equipment, or stakeholders.
- 6.94 For large centralised power generation systems and distributed systems with a significant collective rating operating in parallel with the primary power supply, the following technical studies shall be completed prior to finalising the design to ensure that the system does not adversely impact Defence operations:
- a. load flow study: including assessment of power quality and thermal loading
 - b. fault level study: including evaluation of impacts upon on existing equipment
 - c. protection study: to confirm adequate system protection measures
 - d. earthing study: to ensure compliance with safety and operational requirements.
 - e. additional studies as required by the local NSP.

Protection systems

6.95 All power generating units shall be provided with protection systems commensurate with their capacity, type and operational circumstances. In particular, the following protection shall be provided:

- a. protection to safely disconnect the equipment in the event of a fault event in such a way that minimises damage to the equipment
- b. synchronism check where supplies from different un-synchronised sources can be connected together
- c. anti-islanding protection:
 - i. to requirements of the NSP where generating units run in parallel with the mains
 - ii. where the generating unit is not designed to run islanded.

Microgrid

- 6.96 Microgrids are localised embedded energy systems equipped with control capabilities that allow disconnection from the main utility energy supply and autonomous operation, a mode commonly referred to as “islanding”.
- 6.97 The establishment of a microgrid that manages or controls Defence assets shall require prior approval of the base EOA and the Directorate of Estate Engineering Policy (DEEP).
- 6.98 Defence bases may operate multiple power sources. Depending on the power system configuration, power generation systems can be integrated and controlled within the Defence electrical distribution system through the implementation of a microgrid.
- 6.99 Where a microgrid exercises control over Defence assets, the microgrid controller shall remain under the direct and dedicated control of the EOA.

ELECTRICAL DISTRIBUTION NETWORKS

- 6.100 The Defence electrical distribution system is defined as any electrical power distribution system that is owned, operated and/or maintained by Defence. Defence holds full responsibility for the management of this system and liable for any issues or impacts arising from its operation.
- 6.101 The Defence electrical distribution system is deemed to extend from the source/s of electrical supply through to the point of attachment, point of entry or point of supply at each facility on the Defence establishment.
- 6.102 The source of electrical supply can be one or more of the following:
 - a. NSP connection: The agreed point of supply or connection with the NSP for any incoming supply mains
 - b. Defence owned power system: The output from Defence owned systems such as a CPS or CEPS
 - c. Third party owned power systems: the output from third-party owned systems, including renewable energy sources or conventional power generation systems
 - d. Specialist equipment outputs, such as:
 - i. load levelling equipment, including batteries or super capacitors
 - ii. inverter systems used to discharge submarine batteries.
- 6.103 Defence electrical distribution systems include:

- a. Defence owned HV networks from the point of supply or connection with the NSP to the LV terminals of any distribution transformer, including:
 - i. all protection, supervisory, monitoring and control systems associated with the HV networks above
 - ii. any generation, energy storage and power conditioning systems connected and feeding into the HV networks including their protection, supervisory, monitoring, operating and control systems and associated step-up transformers. Typical examples on the Defence networks include CEPS, CPS, CEP, DRUPs, standby/emergency generators, renewable energy generation and storage systems and power factor correction.
- b. the LV supply from the Defence distribution transformer from the LV tails thorough to the point of supply or connection at the facility or building on the Defence site, including:
 - i. all protection, supervisory, monitoring and control systems associated with the LV networks above
 - ii. any generation, energy storage and power conditioning systems connected and feeding into the LV networks including their protection, supervisory, monitoring, operating and control systems. Typical examples on the Defence networks could be but are not limited to: LEGs, MGLB, renewable energy generation and storage systems, power factor correction
 - iii. the LV cable/power line from the distribution substation LV board to the point of connection/point of supply or to the main switchboard of a building or facility is considered part of the LV network.
- c. Defence establishments where the incoming supply from the NSP is LV, from the point of connection or supply with the NSP thorough to the point of supply or connection at the facility or building on the Defence site, including:
 - i. all protection, supervisory, monitoring and control systems associated with the LV networks.
- d. any embedded generation, energy storage and power conditioning systems connected and feeding into the LV networks including their protection, supervisory, monitoring, operating and control systems. Examples on the Defence networks include:
 - i. LEGs
 - ii. MGLBs
 - iii. renewable energy generation and storage systems
 - iv. power factor correction.
- e. Where UPSs are provided for specific capabilities within the LV reticulation system of a facility, these do not form part of the Defence electrical distribution systems.

Support to Defence capability

- 6.104 The Defence electrical distribution system shall provide reliable electricity supply that supports and sustain Defence operations and capability. Operational facilities may require a

higher levels of redundancy and survivability than those typically found outside the Defence sector. To achieve these objectives, the Defence electrical distribution system must:

- a. provide adequate capacity for known loads: New assets shall include spare physical and electrical capacity to accommodate anticipated future development and/or load growth
 - i. typically, new assets shall not exceed 80% utilisation by known loads unless planning considerations dictate otherwise. For equipment rating purposes, known loads shall be increased by 25%
 - ii. projects utilising existing assets may use available spare capacity without additional infrastructure. However, if additional infrastructure is triggered, sufficient spare capacity shall be incorporated.
- b. HV networks shall be designed with flexibility and redundancy down to the distribution substation level. This configuration shall allow rapid reconfiguration of the system to bypass faulty components.
- c. HV secondary distribution shall generally be configured as ring mains to provide redundancy and flexibility. The exceptions are:
 - i. where the load supported by the substation is of low importance and capability will not be compromised by the absence of redundancy
 - ii. where the cost of implementing a ring main is prohibitive relative to the benefit gained, and adequate embedded generation, standby power, or alternative facilities are provided.
- d. standby power shall be provided in accordance with requirements outlined in the standby power section of this policy
- e. power quality shall be maintained within acceptable limits at all times, except during fault events that cause temporary excursions outside limits
- f. Radio Frequency Interference (RFI) and Emissions: Potential RFI or other emissions from power system elements and components shall be assessed when installations are located near RFI-sensitive facilities. Where necessary, select installation sites at an appropriate distance and/or implement suitable shielding and filtration measures.

Survivability and Resilience

- 6.105 Where Defence capability must be maintained following extreme events (i.e. natural disasters), the role of the Defence Electrical Distribution System (DEDS) in supporting that capability shall be assessed, particularly where post-disaster functionality is required.
- 6.106 If the DEDS is essential to sustain current or future capability, the survivability of individual power system elements shall be considered during design and planning. This may exclude more physically vulnerable construction methods, such as kiosk substations, in favour of more robust alternatives like building-type substations.
- 6.107 This requirement applies not only to power system elements that directly supply critical capability but also to upstream components. For example, if substations on a ring main are compromised, power may not reach the facility. Therefore, all substations within the network may need to be constructed to a higher standard to achieve the required level of survivability.

- 6.108 The DEDS shall be operated to optimise overall reliability and energy efficiency. This includes, but is not limited to, the following practices:
- a. Network Open Points: Normal network open points shall be positioned to minimise operational impact in the event of fault. In general, this requires locating open points toward the middle of ring configurations
 - b. Parallel Operation of Main Transformers: Main power transformers may be operated in parallel, subject to NSP operating protocols, provided they have appropriate load-sharing controls and the resulting fault level does not exceed equipment limits. Special attention shall be given to ensuring that increased earth fault levels remain within the design limits of equipment and earthing systems, particularly older installations.

Standardisation of equipment

- 6.109 HV switchgear and associated ancillary equipment shall be standardised as far as practicable to maximise safety and simplify operation and maintenance within each Defence establishment, and where possible, across the region.
- 6.110 Where multiple equipment type exists within a Defence establishment or region, the selection of the site standard shall be determined in consultation with the site EOA, taking into account:
- a. technical suitability for the intended application
 - b. compliance with the requirements of this manual
 - c. operational performance and availability of local technical support
 - d. factors affecting future service life, including spare parts availability and ongoing manufacture.
- 6.111 Equipment other than the site standard can be considered with the agreement of the EOA and DEEP, where a significant quantity of new equipment is introduced and there is a demonstrable financial benefit to Defence.
- 6.112 Existing site standard equipment may be deemed no longer viable if:
- a. it fails to comply in significantly with this Manual
 - b. it presents safety or operability issues that compromise Defence operations or capability
 - c. it is no longer available, spares parts are scarce, or the equipment is scheduled for discontinuation.
- 6.113 Where non- standard equipment is used or the existing site standard becomes obsolete, a new site standard shall be established using an alternative conformance-based specification. The final decision shall be made in consultation with the site EOA, considering operational and maintenance requirements.

NETWORK AUTOMATION, COMMUNICATION, CONTROL & MONITORING SYSTEMS

- 6.114 The status of the Defence electrical distribution system shall be continuously monitored by the Electrical Operating Authority to ensure that system components do not become overloaded or operate outside their rated conditions. Monitoring shall be achieved through automated systems such as the Power Control and Monitoring System (PCMS) or other approved methods.

- 6.115 To enhance operational reliability and efficiency, the monitoring system shall incorporate Supervisory Control and Data Acquisition (SCADA) functionality, including:
- a. Real-Time Data Acquisition: Continuous collection of voltage, current, frequency, and power quality metrics across the network
 - b. Automated Alarms and Notifications: Immediate alerts for abnormal conditions, overloads, or faults
 - c. Remote Control Capability: Secure remote operation of switches, breakers, and other control devices to enable rapid reconfiguration and fault isolation
 - d. Integration with Network Automation: Support for automated load balancing, fault restoration, and energy optimisation strategies
 - e. Cybersecurity Compliance: All SCADA and automation systems shall comply with Defence cybersecurity standards to prevent unauthorised access or control.

Communication networks

- 6.116 Communication networks associated with Defence control and monitoring systems are generally:
- a. Process networks: these are isolated communications networks that are dedicated to a particular purpose, such as an individual control system, and are generally confined to within a single facility
 - b. Wide area networks: these are communications networks that extend over a large geographic area, although in a Defence context are generally restricted to a single establishment.
- 6.117 To maximise future supportability, communications protocols shall generally use open standards, such as Modbus or IEC 61850. Proprietary protocols are acceptable where the required functionality cannot be achieved otherwise.
- 6.118 Process networks are generally isolated networks not connected to other networks. However, where process networks are bridged onto other process networks or wide area networks this shall be done in such a way that it does not compromise the security of either network.

Defence Engineering Services Network (DESN)

- 6.119 To avoid the proliferation of multiple wide-area networks serving different purposes within an establishment, Defence aims to standardize and consolidate network infrastructure. The objective is to utilise a common wide-area network to support Engineering Services. This standardised infrastructure is designated as the DESN.
- 6.120 The DESN serves as a communications backbone comprising multicore fibre optic cables and associated active equipment. Based on the technical, security and other requirements of each system using the DESN, communications may share fibres within the backbone or utilise dedicated fibres.
- 6.121 The separate systems utilising the DESN may include:
- a. PCMS, including load shedding
 - b. Substation Automation System (SAS).

- c. Generator Control System for large centralised power generation systems
 - d. Building Management System
 - e. UPS Alarms
 - f. fire services
 - g. crash alarms
 - h. site blackout signals
 - i. emergency shower operation alarm
 - j. security access control systems
 - k. closed circuit television
 - l. fuel services SCADA systems
 - m. waste water treatment SCADA systems
 - n. lighting controls
 - o. computer monitored emergency lighting.
- 6.122 The DESN network may provide the dedicated fibre link, using separate cores from those allocated to other services, to support the AGL control system. This link will include nodes at ALERs and Air Traffic Control Tower.
- 6.123 DESN infrastructure must comply with the requirements for an official network as defined in the Defence Communications Cabling Standard and Australian Government Information Security Manual.
- 6.124 Any project that creates, significantly augments or extend systems that requiring the DESN, such as those listed in this section, must:
- a. where an established DESN at the base, integrate the system with the existing DESN, including extending the network if necessary
 - b. where there is not yet established DESN at the base, provide enabling infrastructure within the physical boundaries to allow future connection to the DESN.
- 6.125 When proposing any system that will utilise the DESN for a facility, the design must incorporate DESN infrastructure requirements. This includes provision for racks, entry conduits, cable reticulation pathways, and risers. Where future installation of such systems is anticipated, suitable allowances for DESN infrastructure must be included in the design.
- 6.126 Proposals to omit DESN infrastructure must be submitted to DEEP and DELM for review and endorsement.

Substation Automation System (PCMS, PMS, SCADA)

- 6.127 The SAS is a control and monitoring system that manages all or part of the Defence electrical distribution system within an establishment.
- 6.128 The primary purpose of the SAS is to meet the requirements for remote control of HV switchgear contained in this chapter. Additionally, the SAS provides operator with real-time status information on the HV switchgear under control.

- 6.129 The SAS is applied to perform switching operations within the Defence electrical distribution system, a function that can significantly impact Defence capability. Therefore, cybersecurity must be treated as a critical design and operational requirement. Unless otherwise specified, all SAS implementations shall comply with IEC 62443 Security Level 3 at a minimum.
- 6.130 The SAS is limited to the primary distribution nodes within an establishment. Monitoring and control functions at distribution substations are typically performed by the PCMS.
- 6.131 The functions of the SAS vary between establishments depending upon the configuration of the Defence electrical distribution system. Typical functions can include, as applicable:
- a. control of HV switchgear
 - b. monitoring the status of HV switchgear
 - c. monitoring power flows and other key parameters of the HV network
 - d. monitoring the status of ancillary systems
 - e. performing other control and monitoring functions as required.
- 6.132 The SAS shall employ industrial grade equipment. All equipment installed in substations must be designed and rated for operation in electrically noisy environments to ensure reliability and resilience.
- 6.133 The SAS might monitor status and data that is also required by the PCMS, in which case consideration should be given to avoiding duplication of infrastructure by having the SAS pass data to the PCMS. This shall be done in a manner that maintains the security and integrity of both systems.

PCMS

- 6.134 The PCMS is a supervisory system for the control and monitoring of various systems associated with Defence electrical distribution systems at an establishment.
- 6.135 The functions of the PCMS vary between establishments depending upon the configuration of the Defence electrical distribution system can include, as appropriate:
- a. LV load management in response to load shedding commands from other systems, control of HV switchgear should not be performed by the PCMS
 - b. control and monitoring of Local Emergency Generators across the establishment
 - c. monitoring of HV switchgear
 - d. monitoring of power flows in the HV network and distribution substations
 - e. monitoring of HV secondary equipment e.g. battery chargers and protection equipment
 - f. monitoring of centralised power generation plant or other sources of electrical supply, e.g. CEPS or centralised frequency converters (such as 60 Hz) or renewables
 - g. Defence operational requirements, such as RAAF base blackout systems
 - h. other control and monitoring functions as required.
- 6.136 This policy does not prescribe the type of establishment that requires the implementation of a PCMS nor the extent of the PCMS at an establishment. The determination as to whether a

PCMS is required and its extent shall be determined by assessment of the relevant issues, including but not limited to the following:

- a. existence and capacity of any centralised power generation (either prime or standby) with a Generator Control System that requires the ability to shed remote load
- b. master planning requirements and possible staged implementation
- c. the feasibility/desirability of extending the PCMS to existing facilities at the establishment
- d. operational requirements, such as the extent of operator interfaces (full SCADA system implementation or a simplified HMI)
- e. maintenance requirements, such as data capture, monitoring and upkeep, to enable performance-based unplanned maintenance
- f. consequences of power supply interruption, and the need to manage load to prevent these from occurring, in terms of:
 - g. loss of function
 - h. process disruption
 - i. recovery time
 - j. duration and frequency of power outages.
- k. ability of utility power supply or site emergency power supply to meet the site power requirements
- l. the presence of embedded generation, within the Defence facility able to export electricity back into the NSP network.

6.137 In the design of new (or modifications to) centralised generator plant, HV networks or distribution substations the presumption shall be that, at a minimum, facilities in support future integration with a PCMS shall be provided.

6.138 Generally, PCMS shall be a distributed control system with 3 main components:

- a. PCMS Process Network (PN): A distributed control layer consisting of remote input/output devices (such as PLCs) or remote terminal units (RTU).
- b. SCADA: System that monitors and provides a user interface into the PCMS Process Network
- c. application servers for logging and retrieval of monitored information/data, events and alarms.

6.139 The PCMS shall be rugged and fault tolerant. For instance:

- a. the PCMS shall utilise industrial grade equipment, and equipment used in substations shall be suitable for a noisy electrical environment
- b. the PCMS shall be resistant to cyber-attack. Radio or other wireless links shall only be used following a cybersecurity risk assessment and with Defence approval. Unless otherwise indicated, the PCMS shall comply with at least IEC 62443 Security Level 2

- c. Process Network shall be capable of autonomous operation in the event that communications to the PCMS SCADA are unavailable.
 - d. No automatic control shall be implemented in the PCMS SCADA
 - e. The PCMS fails in a way that minimises impact to operations.
- 6.140 The PCMS SCADA shall provide an operator interface consisting of an operator terminal/s or human machine interface (HMI) to allow functions such as:
- a. configuration of the PCMS PN control functions
 - b. manual control of some PCMS PN control functions
 - c. Monitoring, storing and trending data from the PCMS PN and other systems.
- 6.141 Although the PCMS monitors several mission-critical systems, such as HV electrical network and any CEPS, it shall not control or influence these systems, except for LV load management.
- 6.142 Robust security measures must be implemented to ensure that, in the event of a cyberattack on the PCMS, attackers cannot gain access to mission-critical systems monitored by the PCMS. Interfaces between the PCMS and mission-critical systems shall generally be at a low level. Where high-level interfaces are required, they must be protected using appropriate hardware firewalls.
- 6.143 The DESN shall be designed to withstand cyber security attacks in compliance with Defence and Australian Government Policies, specifically the Information Security Manual and the Defence Security Principles Framework.

HAZARDOUS AREAS

- 6.144 Departures from the state and territory regulations cannot be approved by Defence. Any departures shall be approved by the relevant state or territory regulator prior to consideration of acceptance by Defence.
- 6.145 This manual does not detail the requirements for Defence Fuel management systems or Explosive Ordnance Areas. The policy lead on these topics is Joint Capabilities Group.
- 6.146 This manual does not detail the requirements for LDBO. The policy lead on LDBO is Air Force.

Dossiers

- 6.147 Verification dossiers are required for all hazardous areas and explosives hazardous areas installations.
- 6.148 Restricted electrical area dossiers are required for all Restricted Electrical Area installations.
- 6.149 All new installations shall be provided with a new dossier and all alterations to an existing installation shall be provided with information in a form suitable to amend the relevant dossier in its various forms (digital/printed).
- 6.150 The completed dossiers shall comply with the relevant requirements of AS/NZS 60079.14 and AS/NZS 60079.17 or eDEOP101, and this manual.
- 6.151 HA, EHA and REA dossiers shall include in addition to the content required by the standards and regulations:
- a. Electrical Non-Conformance Dispensations

- b. as-built drawings for the electrical installation inspection and test records including time, date, personnel involved, list of test equipment used, calibration record, measurement method, conditions at time of measurement, results and acceptance criteria. Inspections must verify that the installation is consistent with the design and documentation provided in the dossier
- c. operating/maintenance instructions and records for the operation, maintenance, repair and overhaul of the electrical installations
- d. competency records
- e. lightning protection installation and maintenance records
- f. static control measures and maintenance records
- g. certificates of electrical safety
- h. design and construction certification
- i. for REA dossiers - ex certificates of conformity.

CERTIFICATES OF ELECTRICAL SAFETY

6.152 Electrical Safety certificates (required by the Statutory Electrical Safety Regulator) must be provided to Defence prior to handover of the works.

Defence specific requirements

6.153 The following guidelines articulate Defence specific requirements relevant to this chapter:

- a. 6000 – Defence Electrical Distribution Network Numbering and Labelling
- b. 6001 – Defence Electrical Distribution – Planning Reports
- c. 6002 – High and Low Voltage Distribution
- d. 6003 – Local Emergency Generators
- e. 6004 – Mobile Generator Link Boxes
- f. 6005 – Uninterrupted Power Supplies* Covers AGL
- g. 6006 – Clean Energy Acceleration on the Defence Estate

CHAPTER 7

AERONAUTICAL GROUND LIGHTING (AGL)

GENERAL PRINCIPLE

- 7.1 This chapter applies to land-based aerodromes and heliports. It is intended to support, not replace, the applicable standards. AGL is a safety critical element of aerodrome systems. While inadequate aerodrome design may not affect aircraft airworthiness, it can have a direct and substantial impact on safe flight operations. Consequently, effective planning and correct implementation optimises system efficiency and mitigates operational risks.

Rationale

- 7.2 The rationale of this chapter is to reinforce that all AGL systems must always be in full compliance with all applicable environmental, legislation, regulatory and applicable standards.

Expected outcome

- 7.3 AGL systems shall comply with the Defence Aviation Safety Authority Aerodromes Regulation (DASR 139), Design Manual (DASDRM) and all relevant standards. This chapter also prescribes additional non-aviation safety design requirements necessary to support Defence capability, standardisation, and interoperability with other Defence forces and aircraft operators.
- 7.4 AGL systems shall be designed, installed, altered and maintained following appropriate certification processes compliant with relevant standards and all applicable Defence policies.

AGL system overview

- 7.5 AGL consist of collective lighting systems and visual aids at an aerodrome, providing guidance to aircraft and other users in all conditions, including day, night and low visibility.
- 7.6 AGL systems are a set of interconnected components designed to meet the standards for specific runway categories, such as non-instrument, non-precision instrument, or precision instrument.
- 7.7 Systems include mandatory and advisory lighting for approaches, runways, taxiways, aprons, visual guidance on slopes, aerodrome beacons, Illuminated Wind Direction Indicators, Movement Area Guidance Signs, and associated control and power systems, both normal and emergency use.

Compliance

- 7.8 Compliance with AGL requirements is governed by 2 distinct elements, each with its own regulatory and standards hierarchy:
- a. AGL design and installation:
 - i. governed by DASR 139 – aerodromes. These regulations form the basis for design requirements detailed in the DASDRM
 - ii. supported by this manual and its associated guidelines, which provide technical direction for the implementation of AGL systems.
 - b. AGL electrical installation:
 - i. All electrical aspects of AGL systems must comply with the requirements outlined in the electrical chapter of this manual.

AGL field installation

- 7.9 Guideline 7006 outlines the requirements for AGL field installations, covering topics including, but not limited to:
- a. circuit configuration
 - b. circuit interleaving
 - c. pit and duct system
 - d. primary and secondary cable
 - e. cable joints, plugs and sockets
 - f. series isolation transformers
 - g. luminaire and sign installation
 - h. light fittings (Inset and elevated)
 - i. primary circuit surge protection
 - j. constant current regulators
 - k. tactical landing zone lighting.

Airfield lighting equipment room (ALER)

- 7.10 An ALER holds the control and power equipment for AGL systems. Guideline 7001 outlines the requirements for ALER.

Apron floodlighting

- 7.11 Apron floodlighting may be installed to support low visibility and night-time operations. DASDRM is the primary document for apron flood lighting. Guideline 7004 outlines additional requirements for apron flood lighting.

Distance to run markers (DTRM)/hook cable marker (HCM)

- 7.12 DTRMs and HCMs are Defence specific requirements that provide critical information to pilots about the remaining length of runway available for landing or take-off and the location of the arrester cable.
- 7.13 DASDRM is the primary document for DTRM/HCM. Guideline 7006 outlines additional requirements for DTRM/HCM.

Emergency runway lighting

- 7.14 Defence has specific requirements for emergency runways, which are typically a specified taxiway repurposed as an emergency runway.
- 7.15 The requirement for emergency runway lighting will be determined by aerodrome operator. DASDRM is the primary document for emergency runway lighting. Guideline 7006 outlines additional requirements for emergency runway lighting.

AGL control system (AGLCS)

- 7.16 AGL must be able to be controlled from the ALER and the permanent Air Traffic Control facility. Mobile and/or remote control may be required by the aerodrome operator. Guideline 7002 outlines the requirements for AGLCS.

Compliance by specific design and installation

- 7.17 Some element of AGL will not comply with AS/NZS 3000 part 2. Guideline 7003 outlines the requirements in these circumstances.

Aeronautical ground lighting configuration manuals (AGLCM)

- 7.18 The AGLCM details the AGL configuration layout and outlines the management, control and maintenance requirements for the AGL system on Defence aerodromes.
- 7.19 The AGLCM for a given aerodrome incorporates the following topics:
- a. roles and responsibilities of aerodrome stakeholders
 - b. dispensations applicable to physical characteristics of the aerodrome
 - c. dispensations applicable to the electrical power aspects of permanent aeronautical ground lighting on the aerodrome
 - d. overview of the AGL visual navigation aids provided at the aerodrome
 - e. overview of the aerodrome's ALER(s) and equipment within
 - f. overview of the aerodrome's AGLCS
 - g. details of the aerodrome's AGL maintenance requirements
 - h. details of the aerodrome's AGL visual inspection requirements
 - i. details of the aerodrome's AGL electrical power supply periodic testing requirements
 - j. preventative maintenance schedule and procedures
 - k. spare parts inventory and management system.
- 7.20 The entity that modifies the AGL system is responsible for updating the relevant manual and documentation.
- 7.21 Guideline 7009 provides further information on the AGLCM and associated documentation.

Software Management

- 7.22 All software utilised by the AGL systems shall be maintained to ensure the software is readable and recoverable.
- 7.23 Guideline 7000 provides further information on the software management.

Assurance checks

- 7.24 Assurance activities are outlined in DASDRM (manual of standards MOS part 139).
- 7.25 The aerodromes must be maintained to the approved design configuration to support safe flight operations.
- 7.26 The following activities serve as examples of some mandatory assurance checks:
- a. serviceability inspections
 - b. aerodrome technical inspections
 - c. annual aerodrome manual validation and report.

- 7.27 Additional auditing and compliance checks are to be implemented as recommended in the AGLCM, Service Definition Table and Aerodrome Manual.
- 7.28 Guideline 7000 outlines additional requirements to assurance checks.

Maintenance

- 7.29 AGL systems are to be maintained in accordance with MOS part 139 and the AGLCM.
- 7.30 Any maintenance activity must not change the configuration baseline of the AGL system.
- 7.31 Guideline 7005 and AGLCM includes additional requirements for maintenance activities.

Defence specific requirements

- 7.32 The following guidelines detail Defence specific requirements related to AGL.
- a. 7000 – AGL Design and Installation
 - b. 7001 – Airfield Lighting Equipment Rooms
 - c. 7002 – AGL Control System
 - d. 7003 – AGL Compliance by Specific Design and Installation
 - e. 7004 – Apron Flood Lighting and Test Criteria
 - f. 7005 – AGL Maintenance Practices
 - g. 7006 – AGL General Description and Performance Criteria
 - h. 7007 – Misc. Services (Static) (TBC)
 - i. 7008 – AGL Power Systems (TBC)
 - j. 7009 – AGLCM Template (TBC)

CHAPTER 8

ELECTRIC VEHICLE CHARGING

GENERAL PRINCIPLE

- 8.1 This chapter applies to the installation and use of electric vehicle charging infrastructure (EVCI) on the Defence estate.

Rationale

- 8.2 Deploying electric vehicles (EVs) is a component of Government's Net Zero in Government Operations Strategy, and the Defence Net Zero Strategy which aims to electrify the Defence white fleet, contractor fleet and support vehicles.
- 8.3 To succeed, EVs must be supported by appropriate EVCI.
- 8.4 The benefits of a well-managed EV fleet and associated charging infrastructure include its potential as a distributed energy resource on the Defence estate, improving Defence energy resilience.
- 8.5 Uncontrolled use of EVCI may pose electrical and physical safety risks on the estate, requiring Defence to maintain control over EV charging.

Expected Outcomes

- 8.6 Safe, controlled, and efficient EV charging across the Defence estate.
- 8.7 Integration of EVCI into estate planning and infrastructure projects.
- 8.8 Compliance with Defence EV charging guidelines and hardware specifications.
- 8.9 Support for Defence's net zero objectives through electrification of fleets.

Policy Scope

- 8.10 The EV charging policy is for:
- a. APS and ADF personnel: the parameters for charging EVs with EVCI on the Defence estate
 - b. estate planners and project managers: the circumstances where plans or projects must or may deliver EV charging infrastructure within their plan or project.
- 8.11 Those listed above must review the EV charging guidance documents for the use of chargers and the integration of EVCI into planning and projects. Usage requirements and integration of EVCI in estate planning is summarised below. More detail is in [Guideline 8000 – EV Charging](#).

Use of EV chargers

- 8.12 EV chargers and SPOs intended for EV charging on the Defence estate are primarily for charging of Defence vehicles.
- 8.13 Defence will endeavour to make charging available to Defence members who live on base. In some locations, under the circumstances outlined in the EV charging guideline, EV charging may be made available for use by private vehicles of people who do not live on the Defence estate. The Australian Government charging policy framework requires Defence to adopt a principle of cost recovery when it comes to charging of EVs on the estate.

- 8.14 Charging vehicles via general purpose outlets is not permitted on the Defence estate. This includes any type of power outlet not specifically approved for the charging of electric vehicles.
- 8.15 For further details on the use of EV chargers, refer to Guideline 8000.

Integration of EV charging infrastructure into planning and projects

- 8.16 EVCI must be considered in the Defence infrastructure delivery process.
- 8.17 For estate planning process, the Estate Base Plan (EBP) and Electrical Master Plan (EMP) for each site must be informed by a current Site EV Charging Plan (SEVCP). If there is no EBP or EMP for a site and a SEVCP has not been developed via some other avenue, it falls to individual projects to assess and deliver EVCI.
- 8.18 For Infrastructure project delivery, projects must consider and include the enabling civil and electrical infrastructure to facilitate the installation of EVCI.
- a. If an infrastructure project has car parking of any kind or an electrical upgrade applicable to car parking in scope, the project must include the requirements set out in the EV charging policy in the project scope, using project funding.
 - b. Projects must consider and include charging hardware as appropriate given the known charging need.
 - c. If an infrastructure project must install EV charging hardware under the EV charging policy, the project must include those EV charging hardware requirements in the project scope, using project funding.
- 8.19 For further details on integration of EV charging infrastructure into planning and projects, refer to Guideline 8000.

Charging management

- 8.20 Every EV charger on the Defence estate must be connected to charging management software and must comply with the latest version of the Defence EV charger hardware specification.
- 8.21 Projects installing EVCI must:
- a. ensure all hardware is compatible and enabled with the Defence Charging Management System (CMS)
 - b. ensure network connectivity of chargers and management system to enable monitoring
 - c. commission chargers to the Charging Management Platform as per the EV charger hardware and any associated installation guidance.
- 8.22 For further details on CMS, refer to Guideline 8000.

Defence specific requirements

- 8.23 Guideline 8000 articulates Defence specific requirements relevant to this chapter.
- 8.24 Queries related to EVCI across Defence should be directed to: ev.charging@defence.gov.au.

CHAPTER 9

GLOSSARY

The source for approved Defence terms, definitions and abbreviations is the Australian Defence Glossary (ADG), available on the Defence Protected Network at [ADG - Australian Defence Glossary](#).

Note: The ADG is updated periodically and should be consulted to review any amendments to the data in this glossary.

For the purpose of the policies described in this document, the following definitions apply.

TERMS AND DEFINITIONS

The following list of terms are defined in Defence Instruction – Administrative policy. The definitions are intended to apply to their use in administrative policy framework documents.

Accreditation.

A permission accepted by or given by a state or territory to a person which allows them to lawfully provide services or carry out work in their area of expertise and includes occupational registration or licensing as required in the state or territory where the work is being performed.

Aircraft area.

Any area where aircraft may be parked inside a building that is not a hangar open shelter and any internal adjacent area not divided from that area by fire walls or open space in which fuel vapours can dissipate.

Asset Protection Zone.

An area between an asset and a bushfire hazard where bushfire fuel has been reduced significantly to protect the asset.

Availability (Elec).

The percentage of time a system is available to operate satisfactorily when required to do so. Calculated over a year, or otherwise agreed timescale, as where a system is not always required to operate, the required time to operate excludes the periods it is not required. For example, street lighting systems would exclude daylight hours from the period the system is required to provide the service function. For example, a Central Power System is required always to supply power to a base at a given capacity with an availability of 99.99%. This would mean the system must be able to supply that capacity for 99.99% of a year or, alternatively, be not available for no more than approximately 52.60 minutes over a given year.

Availability (Mech).

The percentage of time a system is available to operate at the required level of performance when required to do so. Calculated over a year, or otherwise agreed timescale, as:

- a. the period a system can provide the service function it is intended to
- b. the period a system is required to provide the service function it is intended to.

For example, a central pumping system is required to provide given capacity at all times with an availability of 99.998%. This would mean the system must be able to supply the required pressure and flow for 99.998% of a year or, alternatively, be not available for no more than approximately 10 minutes over a given year.

Battery Energy Storage System (BESS).

A system that stores energy produced by an energy generator, including chemical or flow batteries.

Building.

Includes a structure and unless the term 'entire building' is used, includes a part of a building.

Building approval.

This is the approval issued by the building surveyor approving proposed building work before construction, demolition or a change-in-use commences including all supporting documentation issued in accordance with the certification process detailed in Chapter 3.

Building assessment.

The assessment undertaken by a building surveyor to confirm that the proposed design and specifications comply with the requirements of the NCC or this manual.

Building connection point.

The point at which the firefighting water enters the building being protected or the associated control valve (if located externally).

Building of type A or B construction.

A building that would be considered to be of type A or B construction applying the deemed-to-satisfy provisions in Section C of NCC Volume One regardless of whether they have been subject to performance solutions relating to fire safety.

Building surveyor.

A person who meets the following criteria:

- a. is a qualified and experienced building surveyor
- b. holds a current accreditation as a building surveyor which allows them to undertake building surveying services in relation to the building work for which they have been engaged
- c. is a building surveyor that holds accreditation in the state or territory where the work is being performed
- d. maintains at least the minimum professional indemnity insurance cover as required by Defence.

Building work.

All construction, refurbishment, alteration and additions in relation to the erection or alteration of a building. Building work includes any of the following:

- a. the installation of a demountable and transportable building that is not a relocated building
- b. building, altering, underpinning, whether by vertical or lateral support, moving a building or element of a building
- c. excavating or filling for, or incidental to, the activities mentioned in paragraph (b) that may adversely affect the stability of a building, whether on the land on which the building is situated or on adjoining land
- d. supporting, whether vertically or laterally, land for activities mentioned in paragraph (b)
- e. plumbing work, drainage work or demolition work
- f. a change-in-use
- g. other work to which either the NCC or Defence policy applies
- h. the preparation of designs and specifications associated with the above work.

Bushfire.

An unplanned fire burning in forest scrub or grassland vegetation, also referred to as a wildfire.

Bushfire attack.

Attack by burning debris, radiant heat or flame generated by a bushfire which might result in ignition and subsequent destruction of a building.

Bushfire hazard.

A fuel complex, defined by volume, type condition, arrangement, and location, that determines the degree of ease of ignition and of resistance to control. *Bushfire hazards* are variable in their severity, with severity levels usually measured in terms of the fire intensity (kW/m) arising from the hazard.

Bushfire prone area.

An area that can support a bushfire or is likely to be subject to bushfire attack including forest, woodland, grassland and scrubland as well as nearby stands of vegetation. Bushfire prone areas do not include maintained lawns, parks or gardens, nature strips, plant nurseries, golf courses, vineyards, orchards or vegetation on land that is used for horticultural purposes.

Capability Manager.

The Capability Managers for Defence are:

- a. the Chief of Army – Land Capability
- b. the Chief of Navy – Maritime Capability
- c. the Chief of Air Force – Aerospace Capability
- d. the Chief Joint Capability Group – Joint Capability
- e. Deputy Secretary Strategic Policy & Intelligence – Strategic Intelligence and Cyber Programs and Geospatial Intelligence and Information and Services.

The Capability Managers for determining the Contribution Factor of a building for Defence are found in the [Defence Capability Manual](#). Refer to the Guidelines for Contribution Factor Forms for further information.

Certificate of completion.

Certificate issued by a building surveyor indicating that a building is substantially complete and occupancy of the building associated with the building approval is permissible.

Note: In relation to a building not on the Defence estate, the term certificate of completion is used in this manual to refer to an occupancy certificate, certificate of occupancy, certificate of classification and the like which is required to be issued in different state and territory jurisdictions.

Certificate of regularisation.

Certificate issued by a building surveyor in accordance with the process in chapter 3.

Note: Chapter 3 applies where the occupancy classification in GEMS is not correct, when the certificate of completion cannot be located, or when a transportable or demountable building is proposed to be relocated.

Certification functions.

The functions carried out by a building surveyor as required by this IEM or other relevant Defence policies including:

- a. issuing of a building approval or variation to a building approval following a building assessment
- b. completing all reports and forms and documenting reasoning and submitting all documents as required by this IEM or Defence policy
- c. assessing whether persons preparing the designs and specifications for proposed building work are competent persons
- d. seeking approval for a performance solution, dispensation or special structures assessment from a Defence policy contact
- e. conducting inspections during building work
- f. assessing inspection reports prepared by others relating to the building approval
- g. issuing a certificate of completion following an assessment of all installation certificates and other relevant documentation which is required by the building surveyor to demonstrate that the building work complies with the building approval, the NCC and relevant Defence policies.

Change-in-use.

A change-in-use means:

- a. an increase in Contribution Factor in a building from CF3, CF4 or CF5 to CF1 or CF2
- b. a change in building occupancy classification.

Note: A change in building occupancy classification occurs when:

- there is a change that results in greater than 10% of the floor area of a storey being used for a different class than its current use e.g. new storage (class 7b) areas in an office (class 5) storey being greater than 10% of that storey
- there is a new laboratory (class 8) in a building of another classification
- there is a new residential sole-occupancy unit (class 2, 3 or 4) in a building of another classification
- there is a change to a sub-classification e.g. class 9b to 9a.

Combustible cladding.

Cladding used on external walls or on ancillary elements attached to external walls that is not non-combustible within the meaning of the NCC and includes but is not limited to aluminium composite panels and expanded polystyrene wall systems.

Competent person (building works).

A person assessed by the building surveyor in accordance with this policy as competent to prepare an aspect of the building design and specification, including a person preparing a performance solution report, because of the individual's accreditation, skill, experience and/or qualifications.

Competent and experienced person.

A competent and experienced person will have been assessed by the delivery agent for the work as having the necessary skill, experience and qualifications. The person must:

- a. have the appropriate registration, accreditation or licencing in the state or territory where the work is to be undertaken
- b. Where registration, accreditation or licencing is not available in the relevant state or territory: hold appropriate registration or accreditation in the relevant area of practice in another Australian state or territory
 - i. comply with the requirements of a registered professional engineer as provided by an appropriate engineering body.

Compliance.

Meeting the requirements of the NCC, relevant standards and legislation.

Conformance.

Meeting the requirements set out in relevant Defence policy and related documents.

Contribution Factor (CF).

Refers to a facility's importance to Defence capability. It is measured on a scale of 1 to 5, with 1 being the greatest rating.

Controller.

The device used to automatically process data and control equipment based upon programmable algorithms as either firmware or software e.g. PLC.

Construction agent.

The party responsible for delivering construction phase activities including all activities related to fabrication, construction and installation and testing and commissioning of new or modified plant and equipment and, where applicable, demolition and removal of plant upon the Defence estate.

Defence electrical distribution systems.

The Defence electrical distribution system is defined as an electrical power distribution system that is owned, operated and/or maintained by Defence.

Defence compliance and conformance requirements.

In relation to:

- a. building works, are the requirements set out in Chapter 3
- b. Defence occupied buildings not on the Defence estate, are the requirements set out in Chapter 3
- c. special structures, are those requirements set out in Chapter 3
- d. performance solutions and dispensations set out in Chapter 3.

Defence estate.

Those properties within Australia and its territories that are owned or leased by the Commonwealth and managed by Defence.

Defence occupied buildings.

Buildings on the Defence estate or leased by Defence that are intended for occupation by Defence personnel, except in Chapter 3 which applies only to Defence-occupied buildings not on the estate.

Delivery agent.

The agent engaged by Defence responsible for implementation of the requirements of this manual and for ensuring contractors, consultants and outsourced service providers provide evidence of *compliance* with the provisions therein. Examples include project management contract administration, management contractor or project delivery services.

Delivery authority.

The applicable Defence directorate responsible for a building project or building work. (Related to Building works).

The applicable Defence directorate responsible for delivery of the project (Related to Fire Engineering).

Demolition authorisation.

Is as required by the Defence Financial Management Manual (FINMAN), being the policy on the demolition or disposal of Defence assets. Schedule 15, Financial Delegations Manual – Delegation to approve the disposal of relevant property lists the positions within Defence that have the authority to approve the demolition or disposal of Defence assets that relate to property, including facilities and infrastructure, and the relevant cost thresholds.

Demolition work.

All work in relation to the demolition of a building.

Demountable buildings.

Buildings constructed in sections in a factory and transported to a site where they are reassembled. Demountable buildings are usually installed on stumps.

Deployable hangar.

A deployable structure for the protection of aircraft and personnel from the weather.

Design.

The intent of the works to be installed by the construction agent in a project in the form of drawings or specifications.

Design Verification.

The review of the design.

Designer.

The party engaged to perform a design function for Defence for a given project.

Design certificate.

The certificate required under Defence contracts to be issued by any competent person engaged to prepare designs or specifications which are required to support the decision to issue a building approval, undertake building work or issue a certificate of completion.

Director of Estate Engineering Policy (DEEP).

DEEP is responsible for administration of this policy. This includes responsibility for ensuring the manual remains current and for monitoring the implementation of the requirements described in the manual.

Dispensation.

An approved variation to Defence policy requirements.

Distortion (individual).

Ratio between the RMS value of an nth order harmonic and the RMS value of the fundamental.

Distribution substation.

A substation used to directly supply loads (typically at 400/230V) consisting of facilities or processes. The distribution substation typically consists of the following major components:

RMU, Transformer and LV Switchboard. However, either or both of the RMU or LV switchboard can be located remote to the substation.

Drainage work.

Any sanitary drainage, liquid trade waste drainage or stormwater drainage system.

Electrical infrastructure.

All electrical equipment on the Defence estate including *Defence electrical distribution systems* and systems within Defence facilities.

Electrical installation.

As defined by the AS/NZS 3000 – Electrical installations “Wiring Rules”.

Electrical Operating Authority.

The agent responsible for the compliance and operational performance of *Defence Electrical Distribution Networks* and *electrical installations* and its systems.

Electric vehicle.

Vehicles powered by electricity (battery electric vehicle – BEV), plug in hybrid vehicles (PHEV), and any other non-specialised or personal transportation equipment requiring charging.

Electric vehicle charging infrastructure (EVCI).

EVCI includes the charging device itself and any ancillary equipment such as mounting poles, concrete pads and bollards that would be installed at the same time as the EV charging device. Also entails civil and electrical enabling infrastructure, which includes but is not limited to electrical and communications conduits, and electrical capacity updates, including space and spare circuit breakers in electrical switchboards for the easy installation of EVCI.

Exempt building work.

Building work identified as exempt at Chapter 3.

Energy.

The capacity to do work, in contrast to electricity, which is a type of energy related to the movement of charged particles. Electricity is produced in a range of processes, primarily by burning fossil fuels (coal, gas) or increasingly from renewable sources (photovoltaics, wind).

Energy security.

The reliability (the ability of a power system to deliver energy in the quantity and quality demanded by energy users) and resilience (ability to prepare for and recover from energy disruptions that impact mission assurance on military facilities) of a base’s energy supply, storage and distribution needed to achieve the desired strategic objective.

Firefighting infrastructure.

All firefighting water equipment on the Defence estate including Defence firefighting water distribution systems and systems within Defence buildings.

Fire safety engineer.

The person who is deemed as being competent in the specific area of practice of fire safety engineering/fire engineering. Accrediting bodies use different names for fire engineers which have the same meaning.

Accreditation or licencing requirements related to building fire safety (other than bushfire matters detailed in Chapter 5) that help determine whether the person is competent include those registered as a fire safety engineer/ fire engineer by the applicable accreditation body in the state or territory where the work is to be undertaken, for example:

- a. Victoria: Building Professional Engineer – Fire Safety by the Victorian Building Commission
- b. Queensland: Registered Professional Engineer in Queensland in the areas of practice of Fire Safety Engineering
- c. NSW: C10 Accredited Certifier Fire Safety Engineering Compliance
- d. Tasmania: Engineer fire safety.

In a state or territory where registration or accreditation for the practice of fire safety engineering is not available, persons must be:

- a. listed on the National Engineers Register (NER) – maintained by Engineers Australia – Fire Safety Engineering
- b. listed on the National Fire Engineers Register (NFER) – maintained by the Institution of Fire Engineers Australia – Fire Engineering
- c. hold appropriate registration/*accreditation* for the area of practice of Fire Safety Engineering/ Fire Engineering in another Australian state or territory.

Performance solutions to building bushfire safety must be undertaken or reviewed by a Fire Protection Association (FPA) Australia Bushfire Planning and Design (BPAD)-Level 3 accredited practitioner.

Fire safety provisions of the NCC.

A reference to the *Fire Safety Provisions of the NCC* is a reference to sections A, C, D (parts D1, D2), E and parts G3, G5 and H1.

Garrison and Estate Management System (GEMS).

A single, integrated system to manage the Defence estate and service delivery and support the Defence estate life cycle. GEMS provides a single source of information for all estate management activities.

Generator Control System (GCS) Station.

The master controller in the GCS Controller network. It performs overall power station coordination functions.

Hangar.

An aircraft area and any hangar annex.

Hangar annex.

Any part of a hangar that is attached to, but not part of an aircraft area e.g. including but not limited to areas for servicing or repair facilities, building services, plant, fuel storage and working accommodation.

Hangar open shelter.

A covered area that is used solely for parking of aircraft. The shelter must have no internal walls and not less than two opposing unenclosed walls.

Harmonic Analysis.

Harmonic Analysis is the determination of the harmonic content of the load or supply.

Harmonic Distortion (total) THD.

Ratio between the RMS value of all harmonics of a non-sinusoidal alternating periodic value and that of the fundamental.

Harmonics.

Distortions of the sine wave that characterises normal AC current. Harmonics are transmitted into an AC line by non-linear loads (i.e. loads that do not draw power in regular sine waves), such as computers, copiers, FAX machines, and variable-speed motors. Harmonics can cause communication errors and equipment damage.

High voltage (HV).

Means a nominal voltage exceeding 1000 volts AC or exceeding 1500 volts DC.

HV Feeders.

HV cabling from the NSP to the Defence establishment ISS.

HV Rings.

HV cabling on the Defence establishment on which substations and switching stations are connected, but not the Feeders or Interconnectors.

HV Switchroom.

Normally an indoor arrangement provided for HV circuit breakers and HV switchgear panels.

Installation certificate.

The certificate required under Defence contracts or this policy to be issued by any person engaged to undertake any part of the building work or to certify the any completed part of building work to which the certificate applies has been installed in accordance with the building approval and any applicable codes, standards or other requirements.

Indoor substations.

Substations where the various major components are housed within a building, either as a freestanding structure or part of a larger structure.

Interconnector.

A direct intertie HV power cable connection between 2 PSSs or PSS/ ISS/ CEPS, that facilitates power flow used to transfer power between them.

Intake substation.

Substation that forms part of the primary electrical distribution system, to which Incomers and Interconnectors are connected and at which the intake voltage is stepped down to the distribution voltage.

Intake switching station.

See primary switching station.

Kiosk substations.

Stand alone, freestanding substations, where various major components are housed within a metal or composite enclosure. This form of construction is limited to distribution substations.

Legislation and regulations.

See Statutory requirements.

Living-in accommodation (LIA).

A residential building as defined in the NCC, other than a class 1a building. Includes class 1b, 2 and 3 occupancy classifications. Defence has 2 categories of LIA, permanent and transient. Transient accommodation is designated at a site for temporary or ad hoc accommodation.

Note: The above definition is for IEM defined LIA. This does not mean that the use of a class 1a NCC occupancy classification on the Defence estate is not permitted. It means that the IEM has no additional requirements for class 1a dwellings.

Low voltage (LV).

As defined in AS/NZS3000.

Microgrid.

Local embedded energy grids with control capabilities, with the capacity to disconnect from the local utility energy supply and operate autonomously. Also known as 'islanding' mode.

Maintainability.

The ability to retain and restore an item to its functional state when maintenance is performed under stated conditions and using prescribed procedures and resources. Measures include probability to repair within a given time, repair rate, and mean time to repair.

Maintenance agent.

The agent responsible to the Defence maintenance authority for the provision of maintenance services for the electrical installation or system.

Maintenance authority.

The Defence agency accountable for the maintenance of a specific electrical installation or system.

Motor-generator (M-G).

Power systems that use a rotating AC generator to generate the needed output power. A motor-generator that is powered by a battery or a diesel or gas-powered engine when utility power is lost, constitutes a rotary or hybrid UPS.

Network operator.

Means the owner, controller, or operator of an electricity supply network.

Network Service Provider (NSP).

The owner and operator of the assets used to provide mains power in the locality where the Defence establishment is located.

Objective.

The primary recordkeeping system for Defence.

Occupancy classification.

A building occupancy classification is the class given to a building based on its proposed or actual use as defined in part A6 of the NCC.

On-line UPS.

An UPS that continuously powers the load from the inverter under normal operation.

Operations agent.

The agent responsible to the Defence Operations Authority for the provision of operations services for the electrical installation or system.

Operations Authority.

The Defence agency responsible for the operation of the specific electrical installation or system.

Outdoor substations.

Substations where the various major components are outdoors without further enclosure.

Performance solution.

As defined in A2G2 of the NCC – Building Code of Australia Volume One.

Plumbing work.

Includes any water plumbing, roof plumbing, sanitary plumbing system or heating, ventilation and air-conditioning plumbing.

Policy contact.

The nominated contact of the directorate responsible for the policy as listed on the Legislation & Policy page on the Defence Intranet.

Point of Supply.

The point where the distribution system of the NSP and the electrical system of the customer are connected (see AS/NZS 60038).

Power Factor.

The power factor is the mathematical relationship between apparent or effective power, measured in kVA, and real or average power, measured in kW. When the current and voltage are in phase, purely resistive load, the power factor is 1. In a purely reactive load in which voltage and current are 90° out of phase, the power factor is 0.

Power Failure.

See AS 62040.3.

Primary network.

The Defence HV networks comprising the incoming supply from the NSP, the ISS or PSS and CEPS, CPS, CEP, CPCS and interconnectors.

Primary power.

See AS 62040.3.

Primary switching station.

Switching station or substation that forms part of the primary electrical distribution system, to which Incomers and Interconnectors are connected.

Project.

Common industry term for estate product. Any activity (or series of activities) including the planning, design, construction, commissioning and decommissioning of any works that are performed in relation to an estate product providing new infrastructure or replacing existing infrastructure.

Project delivery agent.

See delivery agent.

Project Delivery Authority.

See delivery authority.

Recommended.

Desirable but not mandatory. Similar to '*should*'.

Redundancy.

Refers to the duplication of systems or components such that, in the event of a failure of one of more components, the system is able provide the equivalent service.

Related persons.

Defence considers that multiple service personnel occupying a building are related for the purposes of determining the requirements for sole-occupancy units under the NCC.

Reliability.

The probability that an item will perform as intended for a specified period under a stated set of conditions. Usually measured as a probability, failure rate, or mean time between failures.

Resonance.

Resonance results in voltage surges and transients of several times the supply voltage. PFC connected to a supply can cause resonance between the supply and the PFC capacitors or between other equipment and the PFC capacitors.

Ring main unit (RMU).

RMU, compartmentalised HV switching/ protection panels normally switch/ fuse switch disconnector unit. Switchgear normally used at substations.

Relocated building.

A transportable or demountable building that has been previously installed in a location on the Defence estate and has been moved to another location on the Defence estate, whether on the same site or another site.

Renewable Energy Systems

Energy systems that include either photovoltaic solar or battery energy storage systems for energy generation or storage.

Sag.

A low-voltage condition in which the voltage on one or more phases of AC power falls below 80-85% of the nominal value for more than one cycle (1/50th of a second for 50-cycle AC). Can be caused by ground faults, starting large loads, inadequate power supply, utility switching, utility equipment failure, and lightning. Can cause computer crashes and damage equipment.

Shall.

Is to be interpreted as mandatory.

Should.

Desirable but not mandatory. Similar to '*recommended*'.

Special structure.

A building where Defence has determined that full compliance with the NCC would be inappropriate due to the need for operational training realism. Where this occurs the concessions in Chapter 6 are permitted. The same concessions may be applied to a tent, membrane structure or deployable hangar.

Special structures do not include:

- a. buildings used for general training
- b. buildings used for outdoor training such as obstacle courses
- c. a building that is exempt under Chapter 3.

Standby power.

See AS 62040

Static bypass.

See AS 62040

Static switch.

Synchronous bypass facility. The combination static (semiconductor) and electromagnetic switch that bypasses the UPS to connect the critical load to the bypass power source.

Statutory requirements.

Commonwealth, State and Territory legislation, regulations, codes, guidelines, directives, standards etc. applicable to the works or operations been undertaken.

Storage building.

NCC defined class 7b buildings and includes structures used primarily for storage.

Substation (SS).

A facility in the electrical distribution system at which voltage transformation occurs, i.e. the voltage is changed from one voltage to another. The typical substation generally comprises as a minimum a power transformer. Substations may also include the associated switchboards, however either or both can be located remotely.

Substantial alteration.

An alteration or addition of a building is a substantial alteration if the proposed alteration together with any other alterations completed or permitted within the previous 3 years:

- a. increases the floor area of the building by more than 50% of the original total floor area, as defined in the NCC, of the building
- b. there has been or is a proposed change-in-use of an NCC occupancy classification applying to more than 50% of the original building.

A substantial alteration is not a change-in-use resulting in an increase in Contribution Factor in a building from CF3, CF4 or CF5 to CF1 or CF2.

Substantially complete.

Building work is substantially complete when the following are installed, constructed, completed or provided for as required by the NCC, this policy or the building approval:

- a. all fire safety measures
- b. provision of access and egress
- c. wet areas and waterproofing
- d. sanitary installations
- e. all aspects that relate to health and safety
- f. reticulated water is connected

- g. electricity supply is connected to the building and fitted off
- h. all elements of any additional aspects for consultation and approval by Defence have been addressed as required by Defence.

Sustainability ratings scheme.

A ratings scheme is a system that measures and scores the environmental or energy performance of a building, product or service, using set standards e.g. NABERS or Green Star.

Surge.

A HV condition in which the voltage on one or more phases of AC power exceeds 100% of the nominal value for more than one cycle (1/50th of a second for 50-cycle AC). Can be caused by a rapid load reduction or switching and can damage equipment.

Switching station (SWS).

A facility in the electrical distribution system that is specifically designed for electrical switching. Normally housing HV RMUs.

Systems Project Office (SPO).

Defence agency responsible for the engineering integrity of aircraft systems, subsystems and interfaces

Thermal runaway.

A condition that is caused by a battery charging current that produces more internal heat than the battery can dissipate. This condition ultimately causes cell venting and premature failure.

Transients.

Disturbances to electrical power lasting less than one cycle (1/50th of a second for 50-cycle AC). Also referred to as voltage spikes. Can be caused by lightning strikes, sudden load-shedding on the primary power system, and shutdown of equipment that was drawing an extremely large amount of power, or a general rise in voltage on the primary power system after use by many consumers. Voltage spikes can blow fuses or trigger circuit breakers, destroy electronic circuitry, and corrupt stored data.

Transportable buildings.

A small single storey building manufactured in a factory and transported to a site in one piece. Transportable buildings can be skid mounted or installed on stumps and are generally used to provide temporary accommodation of some kind.

Unfuelled aircraft.

An aircraft whose fuel system has had flammable or combustible liquid removed such that no tank, cell, or piping contains more than 0.5% of its volumetric capacity.

Uninterruptible Power System (UPS).

Refer to AS 62640, an electrical device or system providing an interface between the mains power supply and sensitive loads (computer systems, instrumentation, etc.). The UPS supplies sinusoidal AC power free of disturbances and within strict amplitude and frequency

tolerances. If input power is removed from the UPS, it will continue to supply the load without interruption.

UPS, rotary.

An UPS in which a motor-generator set is used.

UPS, static.

A solid-state UPS relying normally on battery power.

Useful life (battery).

The time over which a battery can deliver a useful amount of power (normally defined as 80% or more of the battery's capacity).

Water operating authority.

The agent responsible for compliance and operational performance of the water infrastructure.

Wiring rules.

AS/NZS 3000 – Electrical installations (known as the Australian and New Zealand Wiring Rules) as applicable at the time.

Workshop.

NCC defined class 8 buildings and includes structures used primarily for engineering and workshop activities.

ACRONYMS

ADDP	Australian Defence Doctrine Publication
ADF	Australian Defence Force
ADFP	Australian Defence Force Publication
AFEG	Australian Fire Engineering Guidelines
AFFF	Aqueous Film Forming Foam
AS/NZS	Australian and New Zealand Standard
AS	Australian Standard
BAL	Bushfire Attack Level
BDA	Battle Damage Assessment
BOWS	Building Occupant Warning System
BPAD	Bushfire Planning and Design
DEEP	Director Estate Engineering Policy
DELM	Director of Estate and Land Management
DEPU	Director of Estate Planning and Upkeep
eDSM	electronic Defence Security Manual
eDEOP101	electronic Defence Explosive Ordnance Publication
EEGIS	Estate Engineering Governance and Integrity System
EO	Explosive Ordnance
EPBC	Environment Protection and Biodiversity Conservation Act 1999
EV	Electric Vehicle
EVCi	Electric Vehicle Charging Infrastructure
EWIS	Emergency Warning and Intercommunication System
FDCIE	Fire Detection Control and Indicating Equipment
NCC	National Construction Code including Building Code of Australia – Volumes One and Two and Plumbing Code of Australia Volume Three
NFPA	National Fire Protection Association (USA)
NER	National Engineers Register
NZS	New Zealand Standard
PBDB	Performance Based Design Brief
PPP	Public Private Partnership
PFOS	Perfluorooctanyl sulfonate
SES-Band 1	Senior Executive Service – Band 1
WHS	Work Health and Safety
NFPA	National Fire Protection Association (USA)

